NeuralFieldManifold: Reconstruction of LFP manifold with Lag Embedding

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Abstract

Manifold geometry has become a powerful framework for understanding how the brain organizes high-dimensional neural activity into compact, interpretable structure. To date, however, most theoretical and methodological advances in neural manifold analysis have focused on spiking activity, leaving unresolved a fundamental question: what is the intrinsic geometric structure of local field potentials (LFPs)—population-level signals that reflect coordinated neural dynamics and are widely translatable in clinical and brain–computer interface settings. This question is challenging because standard population-manifold reconstruction methods rely on assumptions that are violated by LFPs, including strong temporal autocorrelation across channels. Here, we show that these very properties enable an analytical solution. Modeling LFPs as bounded autoregressive dynamical systems, we derive an explicit analytical characterization of their intrinsic geometry and prove that each sustained oscillatory component contributes an independent topological degree of freedom. As a result, the lag-embedded dynamics lie on a low-dimensional K-torus, where K is determined by the number of distinct oscillatory modes present in the signal. Aperiodic 1/f-like components and stochastic variability do not alter this topology, contributing only geometric thickness around the manifold. To handle nonstationarity and deviations from fixed linear dynamics in real data, we further introduce a geometry-aware learning network that parameterizes time-varying autoregressive operators, enabling robust recovery of the predicted manifold structure under realistic conditions. We validate these predictions using synthetic systems with known dynamics and demonstrate that the predicted toroidal geometry is recoverable in practice via lag embedding. Applying the framework to real neural recordings, we demonstrate its applicability across multiple, independent electrophysiological datasets forming a cross-species evaluation, with consistent recovery of the predicted manifold structure. Together, these results position intrinsic manifold reconstruction as a framework for probing the dynamical systems underlying local field potentials, with implications for mechanistic modeling and behavioral inference.

NeuralFieldManifold: Tutorials and Code

1. Introduction

Advances in large-scale neural recording technologies have fundamentally reshaped systems neuroscience, producing high-dimensional datasets that challenge traditional single-neuron analyses. A widely adopted framework for interpreting such data is the *manifold hypothesis*, which posits that despite the large number of recorded neurons, population activity evolves along a low-dimensional geometric structure embedded in the high-dimensional observation space (Cunningham and Yu, 2014). From this perspective, neural computation is understood as coordinated population-level dynamics unfolding along intrinsic manifolds, rather than as the independent activity of individual neurons (Vyas et al., 2020). Substantial theoretical and methodological progress has supported this view, providing principled tools for extracting low-dimensional structure from population recordings (Gao et al., 2017). In particular, methods such as Gaussian Process Factor Analysis enable the recovery of smooth neural trajectories while preserving trial-to-trial variability and temporal structure (Yu et al., 2009). Together, these approaches have established geometric analysis as a central paradigm for studying neural population dynamics.

While most existing manifold-based studies have focused on spiking activity (Levy et al., 2023; Chung et al., 2016a; Cohen et al., 2020a; Chung and Abbott, 2021), another fundamental neural signal—the local field potential (LFP)—offers a complementary view of population-level dynamics (Jalaldoust and Zabeh, 2025). LFPs primarily reflect aggregated synaptic transmembrane currents within a local neuronal population, providing a measure of shared input and subthreshold

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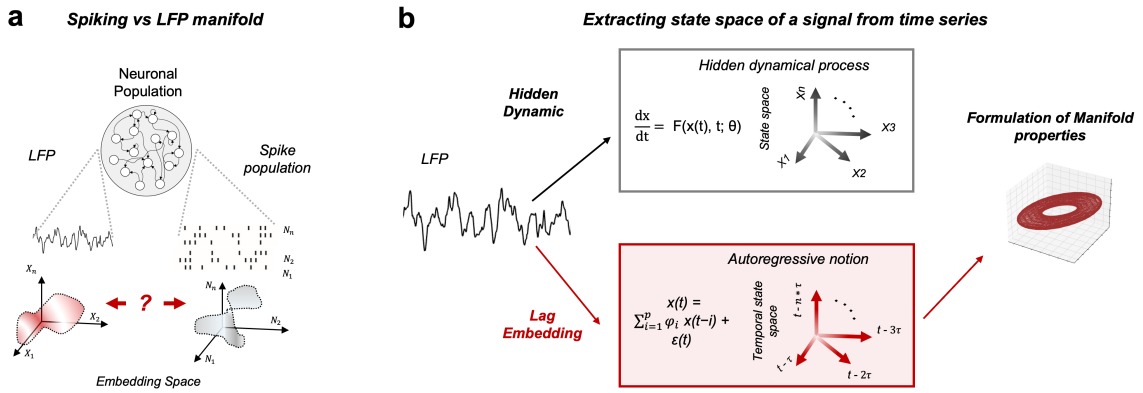


Figure 1. **Conceptual framework for deriving LFP manifold geometry. (a) left** While neural manifold analyses have successfully characterized the geometry of spiking activity, the corresponding geometric structure of aggregate LFP signals remains largely unexplored. **(a) right** Spectral ambiguity in synthetic signals with multiple oscillatory modes: nearby frequencies can yield nearly indistinguishable power spectra under noise, while lag-embedded representations preserve clear geometric structure. **(b)** Our approach models the LFP time series directly using an autoregressive (AR) formulation. Lag embedding of the resulting dynamics enables state-space reconstruction and allows us to analytically characterize the topology of the underlying manifold, which we show to be a k-torus.

processing rather than spiking output (Buzsáki et al., 2012; Einevoll et al., 2019). The biophysical origins of LFPs have been extensively investigated, establishing them as rich but complex signals that capture cooperative neural activity across spatial and temporal scales. Importantly, oscillatory components of the LFP have been repeatedly linked to cognitive functions such as attention, working memory, and cognitive control, suggesting that LFP dynamics carry functionally meaningful computational structure (Miller et al., 2018; Siegel et al., 2012; Zabeh et al., 2023). These observations suggest that, like spiking activity, LFP signals should admit a low-dimensional geometric representation. Specifically, LFPs reflect spatially coherent synaptic currents shaped by recurrent connectivity and shared input (Buzsáki et al., 2012; Einevoll et al., 2019). Recurrent cortical circuits generate collective dynamical modes—such as oscillations and traveling waves—that restrict activity to a small set of dominant degrees of freedom despite high microscopic dimensionality (Churchland et al., 2012; Gao et al., 2017; Muller et al., 2018). Because the LFP integrates across local populations, it primarily captures these coherent modes, and thus is expected to evolve along a low-dimensional manifold defined by the circuit's dominant eigenmodes.

Despite their importance, the intrinsic geometry and dimensionality of LFP signals remain poorly characterized. A central challenge is that commonly used spectral descriptors, such as the power spectral density (PSD), are fundamentally non-identifying beyond restricted classes of linear Gaussian processes: distinct dynamical systems—including quasiperiodic, chaotic, stochastic, and burst-driven processes—can exhibit nearly indistinguishable power spectra. Because spectral statistics retain only second-order structure and discard phase relationships, they are insufficient to recover the underlying state-space geometry or generative dynamics. For example, when oscillatory modes lie close in frequency, moderate noise can merge spectral peaks and obscure distinct dynamical components. Manifold representations, in contrast, can recover the geometric structure of the underlying dynamics under suitable conditions (Gedankien et al., 2025). Yet a principled connection remains missing: when and why does LFP spectral structure imply a specific manifold dimension or topology in delay coordinates?

In this work, we address this gap by developing an analytical framework for the geometry of LFP signals. Our approach leverages two well-established empirical properties of LFP recordings: bounded signal amplitude and a characteristic power spectrum consisting of narrowband oscillatory peaks superimposed on an aperiodic $1/f$ background (Donoghue et al., 2020; Buzsáki and Draguhn, 2004). To move beyond descriptive spectral analyses, we adopt a modeling perspective rooted in stochastic dynamical systems. Continuous-time stochastic differential equations provide canonical models for noisy physical systems (Gardiner, 1985); their discrete-time counterparts are autoregressive (AR) processes, which offer a principled and tractable representation of temporal dependencies in time series data (Stoica et al., 2005; Kaminski and Blinowska, 1991). By modeling the LFP as a (time-varying) AR process, we obtain a generative description of its dynamics without requiring explicit knowledge of underlying biophysical parameters. This provides a first-principles explanation for why highly variable LFP recordings nevertheless concentrate on a bounded intrinsic manifold in delay coordinates.

This formulation enables a precise characterization of the geometry induced by lag embeddings of LFP signals. We show

Work	Data	Goal	Metric(s)	Topology claim?
Takens (1981) (Takens, 2006)	DynSys	Emb. thm.	–	–
Sauer et al. (1991) (Sauer et al., 1991)	DynSys	Embedology	–	–
Ostrow et al. (ICML'24) (Ostrow et al., 2024)	Synth	Seq. repr. $\approx$ delay emb.	Pred. err.	–
Raut et al. (Nature'25) (Raut et al., 2025)	Pupil$\rightarrow$brain	Latent mfld + pred.	R^2, ρ	–
Gonzalez et al. (JCN'24) (Gonzalez et al., 2024)	Hipp. LFP	Geom. coords	Task-dep.	–
This work	**LFP**	**Osc.** $\Rightarrow \mathbb{T}^K$	–	**K-tours**

Table 1. **Delay-embedding related work.** DynSys: dynamical systems; Synth: synthetic; Emb.: embedding; mfld: manifold; pred.: prediction).

that when LFP dynamics are dominated by oscillatory components, the reconstructed state space admits a toroidal structure: each sustained oscillatory mode contributes a circular factor, yielding a k-torus whose intrinsic dimension is determined by the number of dominant oscillations. In contrast, the aperiodic $1/f$ component contributes primarily to the thickness of the embedding around the manifold without altering its essential topology. Beyond analysis, we introduce a reconstruction method that identifies this manifold from data by learning time-varying AR structure and estimating the effective order p—a nontrivial problem under nonstationarity, where correct order recovery is tightly coupled to accurate coefficient estimation. We validate these analytical predictions using synthetic benchmarks with known dynamics and demonstrate their practical relevance by recovering the predicted manifold structure from real LFP recordings. Together, these results establish a principled link between spectral structure and manifold topology, providing a geometric foundation for the analysis of population-level neural signals.

1.1. Relevant work

Time-delay (lag) embeddings are a classical tool for reconstructing latent dynamics from partial observations. Takens' theorem (Takens, 2006) and subsequent work in embedology (Sauer et al., 1991) establish that, under generic conditions, delay-coordinate maps can embed the underlying state manifold, providing a principled foundation for manifold reconstruction from time series. Recent machine learning work has revisited this perspective by analyzing how modern sequence models recover hidden state from history through mechanisms related to delay-coordinate reconstructions (Ostrow et al., 2024). In neuroscience, delay embeddings have been used as practical reconstruction tools for extracting low-dimensional manifolds from biological signals and linking them to brain-wide dynamics; for example, Raut et al. (Raut et al., 2025) reconstruct a latent manifold from a delay embedding of pupil diameter and use it to predict spatiotemporal neural activity, and Gonzalez et al. (Gonzalez et al., 2024) apply diffusion-mapped delay coordinates to hippocampal LFPs to reveal interpretable geometric structure in electrophysiological recordings. Recent deep learning approaches (e.g., sequential autoencoders, neural state-space models, neural ODEs) learn low-dimensional latent dynamics from neural data by optimizing predictive or reconstruction objectives, but the resulting representations are typically implicit and do not explicitly identify geometric or topological structure. In contrast, our approach directly targets topology induced by oscillatory dynamics, linking spectral structure to identifiable manifold geometry.

While these lines of work validate the usefulness of lag embeddings and history-based representations, they largely treat delay coordinates as generic reconstruction devices and do not provide an explicit mapping from oscillatory spectral structure to manifold topology. Our work complements this literature by analytically characterizing the topology induced by sustained oscillatory components in lag-embedded LFP signals and validating the resulting torus dimension predictions on real recordings in a windowed, locally stationary regime.

Why neural manifolds matter. Neural circuits (and artificial neural networks) compute by constructing and manipulating high-dimensional, distributed population representations (Saxena and Cunningham, 2019; Yuste, 2015; Eichenbaum, 2018; Rumelhart et al., 1986). In this regime, a mechanistic account of cognition must explain how information is *embedded and transformed* in population state space—not merely how single neurons "tune"—because population activity often organizes into manifold-like structures (e.g., lines, surfaces, trajectories, and subspaces) whose geometry constrains readout and generalization (Seung and Lee, 2000; DiCarlo and Cox, 2007; DiCarlo et al., 2012). **Geometric analysis turns representational questions into falsifiable, computation-linked quantities**: for example, the untangling program formalizes invariant recognition as increasing linear separability through representation changes (DiCarlo and Cox, 2007; DiCarlo et al., 2012), and manifold-capacity theory connects separability to explicit geometric descriptors such as manifold radius and dimension (Chung et al., 2018; 2016b; Cohen et al., 2020b; Stephenson et al., 2019). This population-geometric framing is especially necessary as neuroscience has moved from classical neuron-centric doctrine to population coding

(Ramón y Cajal, 1995; Sherrington, 1906; Seung and Sompolinsky, 1993; Brunel and Nadal, 1998) and has confronted ubiquitous mixed selectivity in complex tasks (Rigotti et al., 2013). Critically, there remains a *methods gap* when extending manifold reasoning across modalities and datasets: real neural "manifolds" are often noisy and sparsely sampled, and widely used nonlinear embeddings can break under nontrivial topology (Balasubramanian and Schwartz, 2002; Roweis, 2000; van der Maaten and Hinton, 2008; Borg and Groenen, 2005; Gigante et al., 2019; McInnes et al., 2018). This motivates topology- and dynamics-aware approaches—e.g., SPUD (via persistent homology) to recover ring structure in head-direction representations (Chaudhuri et al., 2019; Giusti et al., 2015) and MIND-style manifold inference from transition structure in hippocampal dynamics (Low et al., 2018)—which together make population geometry a principled bridge from large-scale recordings to interpretable computations and generalizable readouts (Nieh et al., 2021).

2. Analytical formulation of LFP as AR process

LFP signals evolve smoothly in time, remain bounded in amplitude, exhibit strong short–lag autocorrelation, and often contain rhythmic structure. In particular, successive LFP samples are highly correlated over short temporal (Einevoll et al., 2013; Buzsáki et al., 2012; Lega et al., 2012; Miller et al., 2009) and spatial lags (Jung et al., 2025; Zabeh et al., 2023; Muller et al., 2018; Das et al., 2025), indicating that much of the signal is predictable from its recent past. This motivates a finite–memory dynamical description: the next sample depends on recent history, so trajectories concentrate on a structured subset of the ambient space rather than filling $\mathbb{R}^N$. Random inputs and measurement noise add thickness around this set but do not determine its shape.

Among possible dynamical models, autoregressive (AR) processes are especially appealing. They provide a parsimonious representation of short–memory temporal dependence, are easy to estimate in sliding windows, and naturally capture both smooth aperiodic ($1/f^{\alpha}$–like) backgrounds and narrowband oscillatory components through their pole structure. This makes AR models well suited for modeling LFP recordings in nonstationary settings.

Within this framework, we formalize how oscillatory structure in neural time series induces low–dimensional geometry under delay embedding. The key result is that persistent oscillations correspond to circular phase variables, whose Cartesian product forms a torus. Noise and aperiodic structure affect the *thickness* and conditioning of the embedding but not its intrinsic topology. Full technical details and proofs are provided in Appendix A.

2.1. Autoregressive view of LFP dynamics

We model LFP activity within a general multichannel autoregressive framework. For clarity, we first present the single-channel form,

$$x(t) = \sum_{m=1}^{p} \phi_m \, x(t-m) + \varepsilon(t), \tag{1}$$

where $\varepsilon(t)$ is bounded, zero-mean noise. The full multichannel formulation—capturing cross-channel interactions through a block-structured autoregressive operator—is provided in Appendix A. Importantly, while our computational method can be applied to individual channels, the modeling framework itself is defined at the multichannel level.

This formulation isolates the deterministic temporal structure—encoded by the coefficients $\{\phi_m\}$—from stochastic variability, which will play only a secondary geometric role.

2.2. Oscillatory modes and poles

In the absence of stochastic perturbations, the asymptotic dynamics of (1) are determined by the roots of its characteristic polynomial. Complex-conjugate roots on the unit circle, $z = e^{\pm i\omega}$, generate sustained oscillations at frequency ω, whereas roots strictly inside the unit circle correspond to damped modes.

Each sustained oscillatory mode introduces a phase variable evolving on a circle. With one oscillation, a delay embedding traces a deformed circle; with two independent oscillations, the trajectory lies on the product of two circles—a 2-torus. In general, K distinct oscillatory modes generate a K-torus.

Proposition 2.1 (Informal)**.** *Assume a bounded LFP signal which, over a local window, contains exactly K distinct persistent oscillatory components (with distinct frequencies modulo π). In the idealized noiseless and marginally stable limit—corresponding to undamped oscillatory modes—a delay embedding using two lags per oscillation lies on a smooth K-dimensional torus. In real data, damping, aperiodic structure, and noise do not introduce additional circular factors;*

instead, they appear as geometric thickness and anisotropy around the underlying torus.

Formal statements and proofs are given in Appendix A.

2.3. Spectral interpretation

For an AR process, the power spectral density (PSD) exhibits sharp peaks at frequencies corresponding to near-unit oscillatory poles, superimposed on a broadband background. Thus, spectral peaks identify sustained oscillatory modes, while the aperiodic $1/f^{\alpha}$ component reflects stable, non-oscillatory dynamics. This establishes a direct correspondence:

Spectral peaks $\leftrightarrow$ k *oscillatory poles* $\leftrightarrow$ k *torus*.

Importantly, distinct dynamical systems can share nearly identical PSDs while differing in phase structure, delay embedding resolves this ambiguity by exposing the underlying geometry.

2.4. Noise, aperiodic structure, and conditioning

Noise and aperiodic components do not alter the intrinsic topology of the embedding. Instead, they control its geometric conditioning: strong amplitude imbalance compresses some circular factors, making them difficult to resolve under noise.

Linear spectral equalization rescales these factors, improving numerical conditioning without changing topology. Thus, equalization reveals existing toroidal structure rather than creating it.

2.5. Analytical prediction

Our analysis yields a concrete geometric prediction for LFP signals under lag embedding. Over short windows, LFPs are well-approximated by stable finite-memory dynamics, making autoregressive (AR) modeling a natural and interpretable surrogate for their strong autocorrelation and bounded amplitude. In this view, the intrinsic manifold dimension is not determined by the ambient embedding dimension, but by the number of *sustained oscillatory modes*: each distinct near-unit complex-conjugate pole pair contributes one independent phase variable, implying that the noise-free lag-embedded trajectory lies on a K-torus $\mathcal{M} \simeq (S^1)^K$, with K equal to the number of distinct oscillatory peaks. In contrast, the aperiodic $1/f^{\alpha}$ component does not introduce new phase variables and therefore does not change the topology; it primarily contributes geometric thickness and poor conditioning, which can obscure weaker torus factors in finite data. This provides a principled advantage over purely spectral methods: while PSDs summarize power, they discard phase information and cannot identify topology, whereas the AR formulation maps spectral structure to poles and poles to explicit manifold geometry. Finally, because real LFPs are nonstationary, identifying this geometry requires time-varying modeling beyond fixed AR fits; this motivates the learning-based formulation introduced next for recovering the predicted manifold structure in realistic settings.

3. Reconstruction of lag embedding for neural data

The analytical results in Section 2 establish a direct connection between oscillatory AR structure and toroidal geometry in delay embeddings. However, the theory relies on a *stationary* AR approximation with fixed coefficients, which is often violated in real LFP recordings. Empirically, LFP signals can exhibit nonstationarity, broadband components, and transient regime changes across short time scales, making fixed linear models insufficient. In this section, we develop a practical reconstruction pipeline that (i) preserves the interpretability of AR-based modeling, (ii) adapts to time-varying dynamics, and (iii) automatically selects the key embedding parameters required for reliable manifold recovery.

Local stationarity bridge. Proposition A.5 assumes fixed AR coefficients; real LFP recordings violate this over long timescales as oscillatory content and amplitudes drift. We therefore operate in a *locally stationary* regime: we model the signal as a time-varying autoregressive (TVAR) process whose coefficient path $\Phi : [0, 1] \to \mathbb{R}^p$ is Lipschitz with constant L_Φ. Under this assumption, one can show that the discrepancy between the TVAR dynamics and a frozen AR surrogate at any reference time t_0 within a window of length w is bounded by $L_\Phi \frac{w}{T} \sqrt{p}\, B$, where B is the local signal amplitude bound (full derivation in Appendix B). Whenever

$$L_\Phi \frac{w}{T} \sqrt{p}\, B \ll 1, \tag{2}$$

the frozen AR surrogate is a valid local approximation. Since polynomial roots vary continuously with their coefficients (Marden, 1966), small coefficient discrepancy implies small perturbation to the pole configuration — and hence, by

Proposition A.5, the lag-embedded trajectory tracks a K-torus determined by the instantaneous oscillatory modes up to a small geometric perturbation, with aperiodic components contributing only thickness around the torus as in Section 2. We verify Condition (2) empirically across all datasets by estimating L_Φ from finite differences of predicted coefficient estimates across adjacent windows and measuring B as the empirical signal amplitude; the condition holds comfortably across all recording sessions, channels, and window lengths considered, confirming that the locally stationary regime is a valid operating assumption rather than a purely theoretical idealization. This provides the bridge from the fixed-coefficient theory of Section 2 to the time-varying reconstruction framework developed below, and directly motivates the DeepLagField architecture, whose smoothness regularizer $\mathcal{L}_{\text{smooth}}$ enforces small L_Φ and whose bounded-energy regularizer $\mathcal{L}_{\text{energy}}$ controls B — together ensuring Condition (2) remains satisfied throughout training.

Setup. Given a univariate LFP segment $\{x(t)\}_{t=1}^{T}$, our goal is to construct a delay embedding

$$\mathbf{z}(t) = \big[x(t),\, x(t+\tau),\, \ldots,\, x(t+(p-1)\tau)\big] \in \mathbb{R}^p,$$

such that the resulting point cloud $\{\mathbf{z}(t)\}$ reveals the intrinsic geometry implied by the oscillatory structure of the signal. This requires selecting the embedding delay τ and the embedding dimension p, and learning a predictive model that is robust to nonstationary temporal structure. Critically, the DeepLagField architecture introduced below directly operationalizes Condition (2): the smoothness regularizer $\mathcal{L}_{\text{smooth}}$ enforces that learned coefficients $\hat{\Phi}(t)$ vary slowly across time, corresponding to a small empirical L_Φ, while the bounded-energy regularizer $\mathcal{L}_{\text{energy}}$ controls B — together ensuring that the locally stationary approximation remains valid throughout training.

Challenges and our approach

Recovering stable geometric structure from neural time series poses several practical challenges. We summarize the key issues and how our approach addresses them.

C1. Local nonstationarity. LFP dynamics are rarely stationary over long horizons; oscillatory content and amplitudes drift across time. We therefore model the signal using a time-varying autoregressive (TVAR) formulation,

$$x(t) \;=\; \sum_{m=1}^{p} \phi_m(t)\, x(t-m) \;+\; \varepsilon(t) \tag{3}$$

which adapts across regimes while preserving an interpretable lag-based structure.

C2. Geometric identifiability. Accurate prediction alone does not guarantee meaningful embeddings: flexible models can fit the signal while producing unstable or misleading geometry. To promote geometric stability, we introduce physics-informed regularization that enforces bounded energy and smooth temporal evolution of $\phi_m(t)$, reflecting empirical properties of LFP recordings and local stationarity.

C3. Delay and order selection. The embedding delay τ and effective order p directly determine the resolved geometry, yet are difficult to tune reliably. We automate this process using average mutual information (AMI) for delay selection (Fraser and Swinney, 1986) and an adaptive order-selection mechanism within the physics-informed model.

Summary. Together, these components yield a reconstruction pipeline that is robust to nonstationarity and explicitly regularized for geometric consistency, enabling reliable recovery of the oscillatory manifolds predicted by our theory.

Network architecture: DeepLagField

We introduce *DeepLagField*, a structured neural model for estimating time-varying autoregressive (TVAR) representations from neural time series. The architecture is designed to produce stable, interpretable lag-based dynamics under nonstationarity, in line with the geometric analysis developed in this work. Figure 2 middle panel summarizes the model. Implementation details, compute configuration, and a small-scale scalability probe are reported in Appendix J.

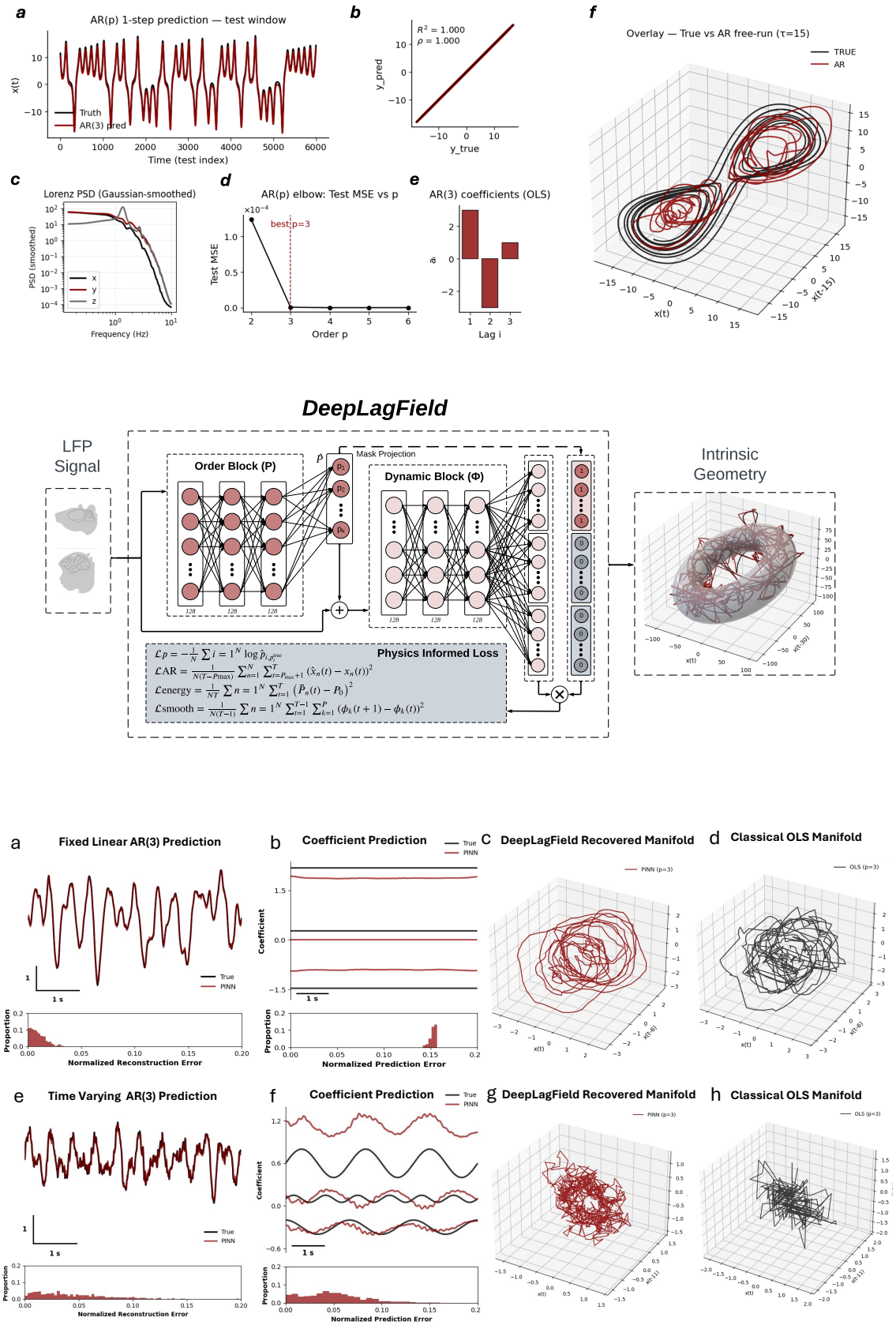


Figure 2. **Integrated analysis pipeline: from synthetic validation to model architecture and PINN evaluation. Top panel (a–f):** Manifold reconstruction of a synthetic Lorenz signal using an $\mathrm{AR}(p)$ model, including prediction accuracy, spectral structure, model selection, coefficient estimation, and delay-embedded manifold recovery. **Middle panel:** DeepLagField architecture, consisting of an order block for lag selection and a dynamic block for estimating time-varying AR coefficients, with physics-informed constraints guiding the learned representation. **Bottom panel (a–h):** PINN-based reconstruction on synthetic LFP-like signals, comparing fixed linear and time-varying AR(3) dynamics, and showing improved signal prediction, coefficient recovery, and preservation of toroidal structure relative to OLS.

Overview. Given an input local field potential (LFP) signal $x(t)$, DeepLagField produces a time-indexed set of autoregressive coefficients $\Phi(t) = [\phi_1(t), \ldots, \phi_p(t)]^\top$, which define a locally stationary TVAR model $x(t) \approx \sum_{m=1}^{p} \phi_m(t)\, x(t-m)$. The architecture consists of two coupled modules: an *order block*, which learns a sparse and interpretable lag support, and a *dynamic block*, which outputs time-varying coefficients conditioned on this support. Physics-informed losses are applied during training to regularize the solution and encode prior knowledge about neural signal structure.

Coefficient prediction. Given a short temporal context window, the network predicts candidate time-varying AR coefficients $\{\phi_m(t)\}_{m=1}^{p_{\max}}$. We implement this mapping using a lightweight feedforward backbone; the specific parameterization is not essential, provided it preserves the lag-based structure required by the theoretical analysis.

Adaptive order selection. Rather than fixing the AR order, DeepLagField incorporates an explicit *order block* that learns a sparse, time-dependent mask over candidate lags. This yields an effective order $\hat{p}(t)$ that adapts across regimes while remaining interpretable. In practice, the learned support aligns with the number of dominant oscillatory modes observed in the spectrum.

Physics-informed regularization. To stabilize learning and avoid geometrically spurious solutions in an overparameterized setting, we augment the reconstruction objective with physics-informed penalties reflecting two empirical properties of LFP signals: bounded short-term energy and smooth temporal evolution of dynamics. These regularizers constrain the solution space without requiring explicit supervision of the coefficients. The exact loss terms and their implementation are provided in Appendix H.

Training and delay selection. The model is trained end-to-end using a differentiable order-selection mechanism. Embedding delays are selected using an average mutual information (AMI) criterion. Full training details and sensitivity analyses are given in Appendix H and Appendix D.

Synthetic evaluation

We evaluate *DeepLagField* on stationary AR and nonstationary TVAR benchmarks with known ground-truth order and coefficients. Synthetic data generation is described in Appendix F. Table 3 reports performance averaged across orders and seeds. While several baselines achieve low one-step prediction error, they exhibit large order errors and unstable estimates of p. This highlights that prediction accuracy alone is insufficient for recovering the underlying dynamical structure.

Table 2. Ablation on TVAR benchmarks. Prediction error alone is insufficient for stable order recovery. Full results in Appendix G. Lower errors are better (↓).

Model	TVAR coeff. error ↓	Pred. MSE ↓	$\Delta p/p$ ↓
Full	0.00439	0.18379	0.30113
No AR(P)	0.00466	0.18348	0.53448
No P	0.65213	0.74777	0.30075
No energy	**0.00376**	0.18459	**0.29885**
No smooth	0.01545	**0.17845**	0.30053

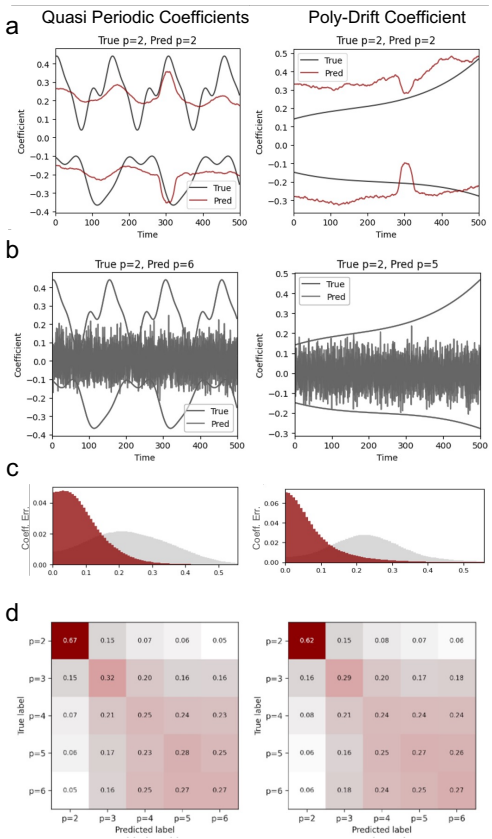


Figure 3. Synthetic data representatives with different coefficient. (a) the coefficient and its prediction. (b) the coefficient before training. (c) the coefficient error before and after training. (d) order recovery confusion matrix.

In contrast, *DeepLagField* substantially reduces coefficient error and recovers the effective order reliably, while maintaining competitive prediction performance. This behavior reflects the intrinsic difficulty of order identification in nonstationary settings and the benefit of physics-informed regularization for learning geometrically meaningful representations.

Ablation: Necessity of geometric constraints. To isolate the functional role of each structural component in LagEmbedDeep, we perform an ablation study on the time-varying AR benchmarks by removing individual constraints while keeping the remaining architecture, optimization procedure, and training data fixed. We evaluate performance using three complementary metrics: (i) mean-squared error (MSE) of the recovered TVAR coefficients, (ii) one-step signal prediction MSE, and (iii) accuracy of autoregressive order recovery averaged across candidate orders.

Table 2 summarizes the results. The full model achieves low prediction error while maintaining stable recovery of both coefficients and effective model order. Since order recovery is evaluated over a heterogeneous, multi-class nonstationary benchmark with multiple candidate orders, accuracy values are most meaningful when interpreted comparatively. Removing the autoregressive structure entirely leads to a collapse in both coefficient estimation and signal reconstruction accuracy, confirming the necessity of explicit AR modeling. Removing temporal smoothness substantially degrades coefficient recovery despite only modest changes in prediction error, highlighting the role of local stationarity in stabilizing nonstationary identification.

Most notably, removing the order-consistency constraint yields a nearly identical signal prediction error to the full model, yet causes a severe degradation in order-recovery accuracy. This demonstrates that predictive accuracy alone is insufficient for identifying intrinsic temporal structure: models can fit the data while failing to recover the correct dynamical order. Together, these results show that the geometric constraints imposed by LagEmbedDeep are essential for disentangling prediction performance from structural identifiability. Full per-order results are reported in Appendix G.

Lorenz stress test (nonlinear out-of-model). As an additional stress test, we evaluate the ability of the reconstruction pipeline to handle strongly nonlinear dynamics using the Lorenz system. While Lorenz dynamics are not intended to model LFPs, they provide a canonical benchmark for testing robustness under chaotic temporal structure. Qualitative embeddings are shown in Figure 2, top panel.

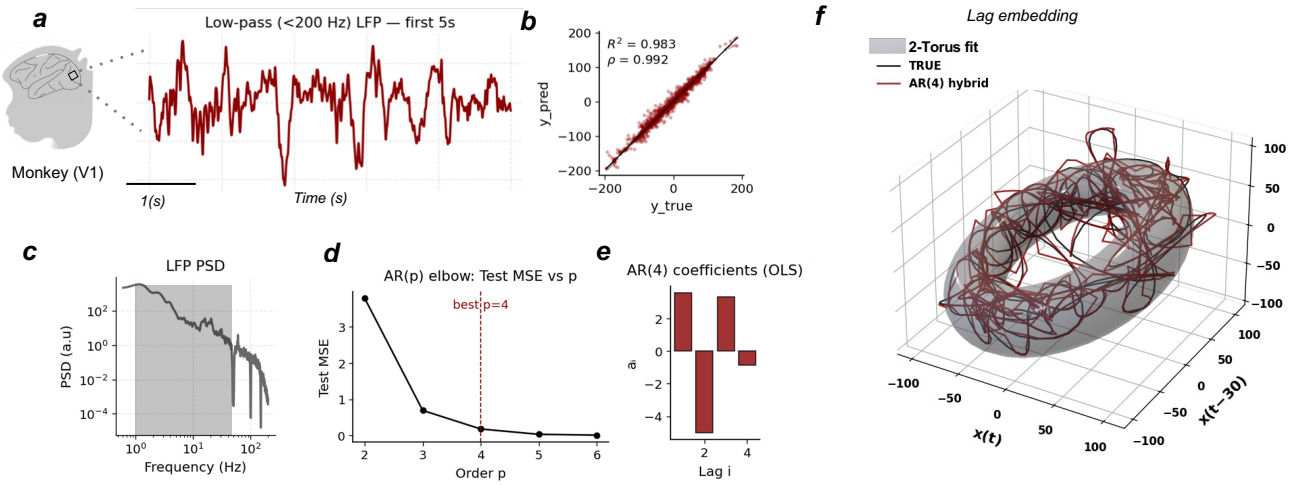


Figure 4. **Manifold reconstruction of monkey (V1) LFP.** (a) Example segment of low-pass filtered (<200 Hz) local field potential from primary visual cortex (V1) of monkey. (b) Prediction accuracy of the fitted AR(p) model on held-out test data: scatter of predicted vs. true samples with R^2 and Pearson ρ confirming strong agreement. (c) Power spectral density (PSD) of the signal, showing $1/f$-like background structure and narrowband oscillatory components. (d) Model-order selection by the "elbow" in test MSE, yielding p=4 as the optimal order. (e) Estimated AR(4) coefficients obtained via ordinary least squares. (f) Delay-embedded trajectory of the LFP and corresponding AR(4) free-run prediction, overlaid with the fitted 2-torus geometry. The embedding reveals that the sustained oscillatory structure of the LFP is well described by a two-circle product manifold, with background variance contributing only thickness around the torus.

Neural data experiments

We next validate LagEmbedDeep on real LFP recordings, where the key question is whether the recovered embeddings reflect the oscillatory organization predicted by spectral structure. We emphasize cross-domain generalization by evaluating across two distinct experimental settings (details in Appendix F).

Datasets and preprocessing. Our primary dataset consists of LFP recordings from primate visual cortex (V1), complemented by rodent hippocampal LFP recordings to assess cross-species generalization. Signals are downsampled to a common sampling rate, notch-filtered to remove line noise, and segmented into epochs aligned to experimental events (full preprocessing and data aggregation details in Appendix F).

Spectral signatures and geometric predictions. Both datasets exhibit a combination of broadband $1/f$-like background activity and narrowband oscillatory peaks. In primate V1, we observe two dominant peaks in the alpha/beta range, suggesting two persistent oscillatory modes. In rodent hippocampus, we observe a dominant theta peak with additional broadband fluctuations. Under the framework of Section 2, these spectra predict a two-torus geometry for the primate recordings and a one-torus geometry for the rodent recordings, with broadband background contributing primarily to thickness rather than topology.

Recovered embeddings and topological summaries. A central question in this work is whether the geometry predicted by our analytical framework can be recovered from real LFP recordings, where signals are noisy, broadband, and only locally stationary. Remarkably, we find that *LagEmbedDeep* consistently recovers low-dimensional embeddings whose topology matches the spectral structure of the underlying recordings across two distinct experimental settings.

In primate V1, the LFP spectrum exhibits two pronounced narrowband oscillatory components on top of a $1/f$-like background (Figure 4).

Our theory predicts that each persistent oscillatory mode contributes an independent circular factor in the delay embedding, yielding a two-torus geometry. Consistent with this prediction, the recovered embedding forms a clear toroidal manifold and is well-approximated by a two-torus fit (Figure 4p). Moreover, the learned reconstruction achieves near-identity agreement between predicted and true samples (Figure 4d), indicating that the recovered geometry is not an artifact of visualization but reflects genuine predictive structure in the signal.

Table 3. Comparison of reconstruction performance on AR and TVAR LFP-like benchmarks. We report mean values averaged across model orders and seeds. Lower is better (↓) for error metrics, higher is better (↑) for stability.

Method	TVAR coeff. error ↓	Pred. MSE ↓	Order error Δp ↓	Order stability ↑
MLP	0.0384	0.1778	N/A	N/A
OLS	0.0040	0.1833	N/A	N/A
Transformer	2.0650	0.0128	N/A	N/A
DeepLagField (without PINN)	0.0134	0.1790	1.1099	0.6997
DeepLagField	0.0044	0.1838	1.1120	0.6989

In rodent hippocampus, the spectrum is dominated by a single sustained oscillatory peak (theta), suggesting a single circular factor and hence a one-torus geometry (Figure 17f). In this case, the recovered embedding collapses to a one-torus manifold, again matching the predicted topology, with additional thickness attributable to broadband fluctuations (Figure 17f). Recovered embedding was also found for rodent cortex EEG signal (Figure 18f), which further validates our theory.

Importantly, the same reconstruction pipeline generalizes across species and brain regions without retuning embedding heuristics, suggesting that *LagEmbedDeep* captures a robust geometric signature of oscillatory organization in neural activity.

Topological quantification and population-level validation. While the lag embeddings in Figures 17–18 provide qualitative evidence for toroidal geometry, we next ask whether this structure can be confirmed rigorously using topological data analysis and whether it is statistically distinguishable from chance. Figure 5 addresses both questions across datasets.

Topological confirmation. We apply persistent homology to the recovered lag embeddings and compute Betti numbers $(\beta_0, \beta_1, \beta_2)$ directly from the point cloud (Figure 5a–b). The persistence diagrams reveal 1 connected component ($\beta_0 = 1$), 2 independent loops ($\beta_1 = 2$), and 1 cavity ($\beta_2 = 1$), matching the expected signature of a 2-torus $(1, 2, 1)$ — consistent with the two dominant oscillatory modes identified spectrally in primate V1. Across the population, $\beta = (1, 2, 1)$ is the dominant configuration (46% of windows), followed by $\beta = (1, 1, 1)$ (40%), with only 14% of windows classified as other, confirming robust recovery of the predicted toroidal topology across recording sessions.

Fitting procedure validation. To verify that the torus fitting procedure is well-calibrated, we evaluate it on synthetic AR(2) and AR(4) signals with known ground-truth geometry (Figure 5c). The recovered torus scores and errors are consistent with the expected topology in both cases — the procedure correctly distinguishes the one-torus geometry of AR(2) from the two-torus geometry of AR(4) — confirming that the quantification is not an artifact of the fitting pipeline but reflects genuine geometric structure in the signal.

Population-level statistical validation. At the population level, torus scores and R^2 values for real monkey LFP recordings are significantly higher than a null distribution obtained from phase-shuffled surrogates (Figure 5d; $p < 0.001$ for all metrics), while torus error is correspondingly lower in real data than in the null. These differences are robust across a wide range of embedding delays τ and window sizes, with performance stabilizing rapidly and remaining stable thereafter, confirming that the recovered geometry reflects genuine oscillatory structure rather than statistical coincidence.

Sensitivity analysis. Sensitivity heatmaps confirm that the two-torus structure in monkey recordings and one-torus structure in mouse recordings are stable across the full range of τ and recording duration examined (Figure 5e), with high torus scores and low torus errors persisting throughout — consistent with the robustness analysis in Appendix E.

Behavioral relevance: sleep stage decoding from manifold geometry. A key question is whether the recovered toroidal geometry carries behaviorally meaningful information beyond what is captured by standard spectral features. We address this using the mouse cortex EEG dataset, where ground-truth sleep stage labels (Wake, NREM, REM) are available, and ask whether geometric parameters of the fitted torus — the major radius R_1, minor radius R_2, and tube radius r — can decode sleep state.

Figure 7a shows that all three torus parameters vary systematically across sleep stages, with particularly pronounced separation in R_2 and r, reflecting changes in the oscillatory structure of cortical dynamics across behavioral states. These parameters are not hand-crafted spectral features but emerge directly from the manifold geometry recovered by our framework.

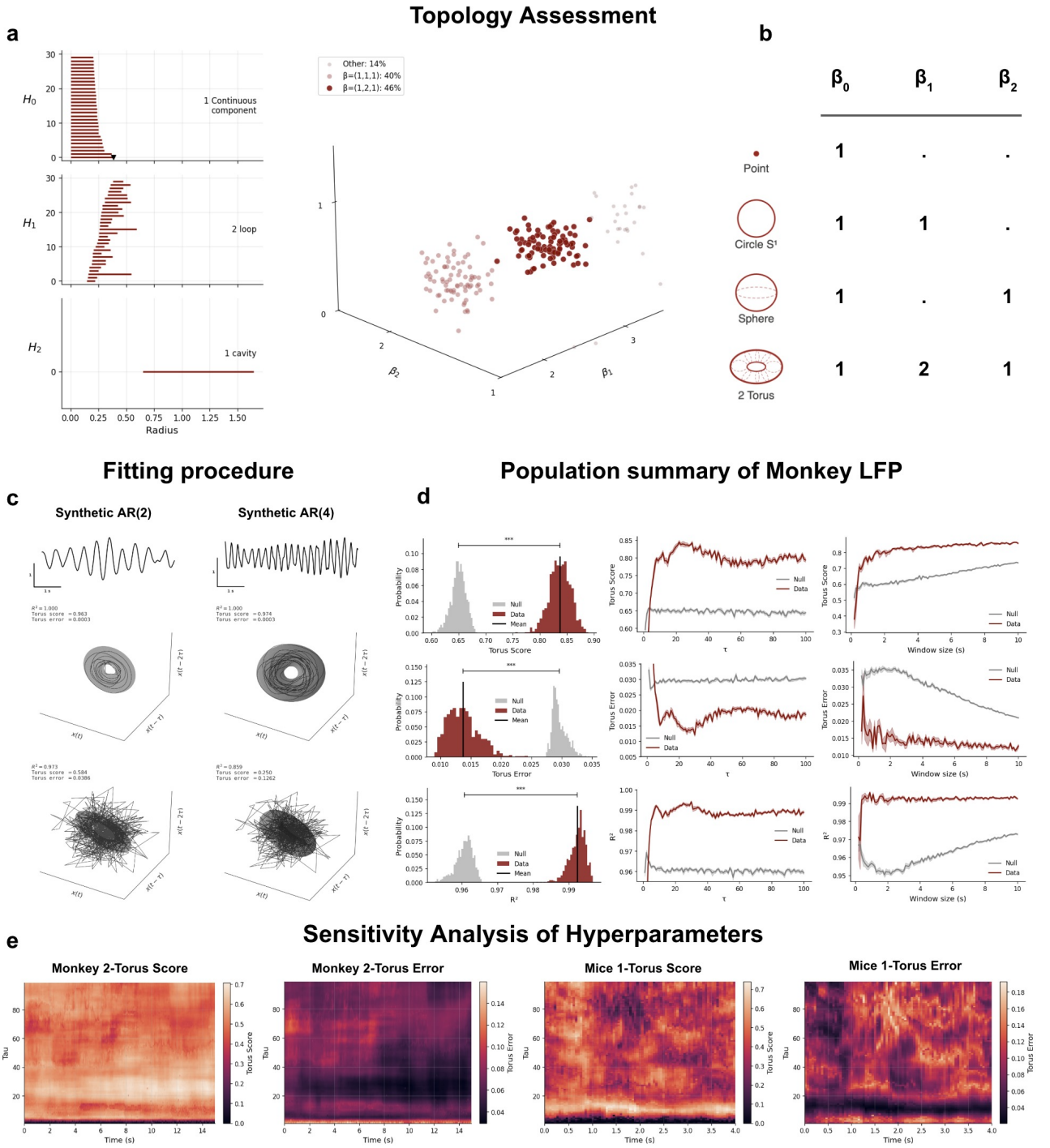


Figure 5. **Topological quantification and population-level validation of recovered manifold geometry. (a)** Persistent homology analysis of the recovered lag embedding from monkey V1 LFP. Persistence diagrams for H_0, H_1, and H_2 reveal 1 connected component, 2 independent loops, and 1 cavity respectively, matching the Betti number signature $(\beta_0, \beta_1, \beta_2) = (1, 2, 1)$ of a 2-torus. The scatter plot shows the distribution of recovered Betti numbers across recording windows: $\beta = (1, 2, 1)$ is the dominant configuration (46%), followed by $\beta = (1, 1, 1)$ (40%), with 14% classified as other. **(b)** Reference table of Betti numbers for canonical topological spaces, confirming that the recovered signature is consistent with a 2-torus rather than simpler geometries. **(c)** Fitting procedure validation on synthetic AR(2) and AR(4) signals with known ground-truth geometry. Top row: lag embeddings with AR model fit (R^2, torus score, torus error annotated). Bottom row: corresponding free-run predictions. The procedure correctly identifies one-torus geometry for AR(2) and two-torus geometry for AR(4), confirming calibration of the quantification pipeline. **(d)** Population-level statistical validation for monkey LFP. Left column: distributions of torus score (top), torus error (middle), and R^2 (bottom) for real data (dark red) versus phase-shuffled null (gray), with means indicated; real data is significantly higher in torus score and R^2 and lower in torus error ($p < 0.001$, all metrics). Center and right columns: sensitivity of torus score, torus error, and R^2 to embedding delay τ (center) and window size (right) for data versus null, showing stable separation across hyperparameter choices. **(e)** Sensitivity analysis heatmaps showing torus score and torus error as a function of τ and recording time for monkey (2-torus) and mouse (1-torus) datasets. High torus scores and low torus errors persist across the full range of τ and duration examined, confirming robustness of the recovered topology to hyperparameter selection.

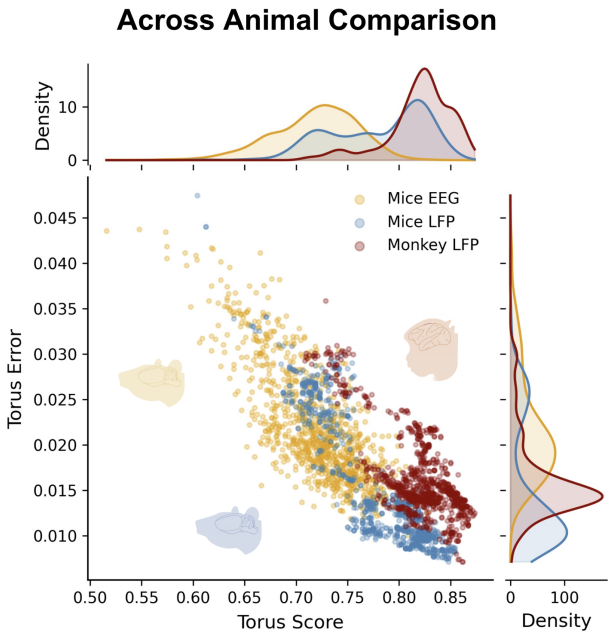


Figure 6. **Cross-species consistency of manifold structure: torus-based quantification across recording modalities. (A)** Joint distribution of torus score and reconstruction error across mouse EEG (gold), mouse LFP (blue), and monkey LFP (dark red). Each point represents a recording segment summarized by the proposed quantification pipeline. All datasets lie along a shared monotonic regime, where higher torus score corresponds to lower reconstruction error, indicating that stronger toroidal structure yields more self-consistent dynamical models. **(B)** Marginal distributions of torus score (top) and reconstruction error (right). Monkey LFP is shifted toward higher torus scores and lower error, consistent with stronger and more stable oscillatory structure. Mouse EEG exhibits broader variability and lower torus scores, reflecting weaker or noisier geometric organization, while mouse LFP occupies an intermediate regime. **(C)** Across species and recording modalities, data collapse onto a common low-dimensional continuum defined by torus score and error. This suggests that toroidal manifold structure is a modality-invariant property of neural field dynamics, and that the proposed metric provides a unified summary of geometry–dynamics consistency across datasets.

Linear discriminant analysis (LDA) on the full set of torus geometric features yields clean separation of sleep states in the projected space (Figure 7c), with decoding accuracy of 94% for NREM, 65% for Wake, and 63% for REM (Figure 7b). Critically, torus features achieve an overall F1 score of 74%, substantially outperforming a band power baseline (60%) across all three sleep stages (Figure 7d) — despite using only three interpretable geometric parameters derived from the manifold structure, compared to the multi-band power features of the baseline. These results demonstrate that the toroidal geometry recovered by our framework is not merely a mathematical curiosity but encodes physiologically and behaviorally relevant structure in neural dynamics.

Geometric validity diagnostic. To verify that the recovered toroidal geometry reflects genuine oscillatory structure rather than fitting artifacts, we apply a Procrustes disparity diagnostic calibrated on synthetic true and spurious tori (full details and figure in Appendix K). Monkey V1 LFP recordings yield a mean Procrustes disparity of 0.1477, above the true torus threshold (0.0820) but below the spurious torus bulk, while mouse EEG recordings yield 0.4092, overlapping with the spurious torus distribution — consistent with the noisier torus geometry observed qualitatively in Figure 18 and confirming that our framework correctly grades geometric quality rather than producing binary pass/fail classifications.

4. Discussion

We presented an analytical framework linking oscillatory structure in local field potentials (LFPs) to low-dimensional geometry. Modeling LFPs as autoregressive processes, we showed that their noise-free dynamics evolve on a K-torus, with intrinsic dimension determined by the number of distinct sustained oscillatory modes. This yields a closed-form connection between spectral structure and manifold geometry, complementing existing empirical approaches to neural manifolds (Cunningham and Yu, 2014; Vyas et al., 2020).

A key implication is the interpretation of the LFP manifold as an *input manifold*. While prior work has largely focused on

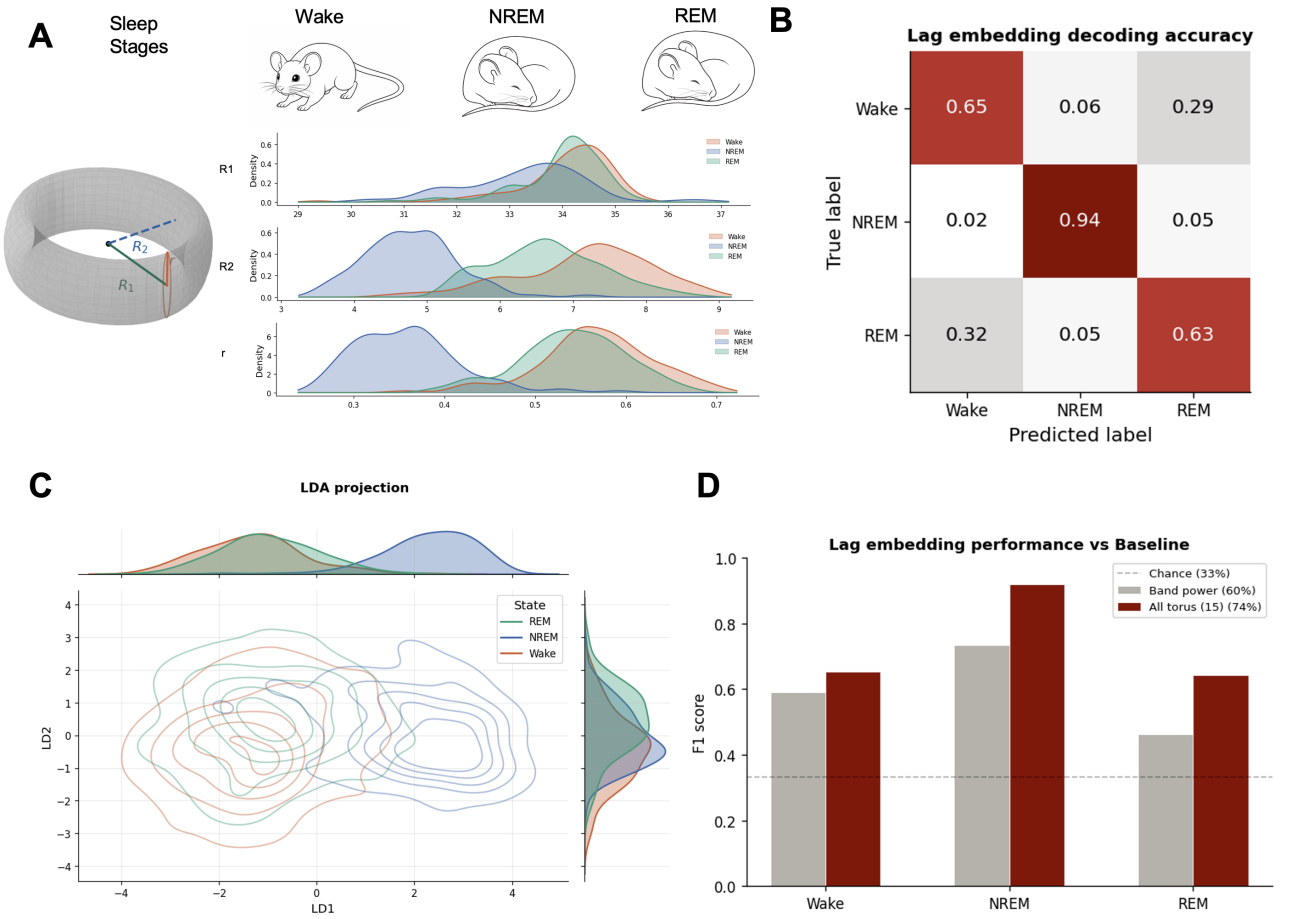


Figure 7. **Behavioral relevance of recovered manifold geometry: sleep stage decoding from torus parameters. (A)** Distributions of torus geometric parameters — major radius R_1, minor radius R_2, and tube radius r — across Wake (pink), NREM (blue), and REM (green) sleep stages recorded from mouse cortex EEG. All three parameters vary systematically across states, with R_2 and r showing particularly pronounced separation. **(B)** Confusion matrix for sleep stage decoding using torus geometric features as input to a linear classifier. Decoding accuracy is 65% for Wake, 94% for NREM, and 63% for REM, substantially above chance (33%). **(C)** LDA projection of torus feature space onto the first two linear discriminants (LD1, LD2), with marginal distributions shown for each sleep stage. The three states form separable clusters in the manifold feature space, confirming that torus geometry encodes behaviorally relevant structure. **(D)** Per-class F1 scores comparing torus geometric features (dark red; 74% overall) against a band power baseline (gray; 60% overall) and chance level (dashed; 33%). Torus features outperform band power across all three sleep stages, demonstrating that manifold geometry carries information beyond standard spectral descriptors.

spike-based manifolds as output representations of neural computation (Vyas et al., 2020), LFPs primarily reflect synaptic and dendritic activity, capturing shared inputs to neural populations (Buzsáki et al., 2012; Einevoll et al., 2013). From this perspective, the toroidal geometry we identify characterizes the structured space of inputs available to a circuit, potentially constraining trajectories on spike-derived manifolds.

Our framework also provides a geometric reinterpretation of neural oscillations. Rather than treating rhythms as isolated frequency bands with distinct functional roles (Buzsáki and Draguhn, 2004; Siegel et al., 2012), sustained oscillations emerge as coordinate directions of the LFP manifold, with multiple rhythms naturally giving rise to higher-dimensional tori.

A final conceptual point is that this modeling choice is not merely a convenient approximation, but reflects a working hypothesis about cortical dynamics: over short temporal scales, population activity is well approximated by linear dynamics, while longer-term changes in behaviorally relevant structure arise through gradual modulation of effective network parameters. The time-varying autoregressive (TVAR) formulation makes this separation explicit by decoupling fast activity dynamics from slow coefficient drift, yielding parameters that map directly to interpretable dynamical quantities such as time-resolved oscillatory frequency, damping, and stability. More general nonlinear autoregressive models can, in principle, represent broader state-dependent transformations, but they often conflate instantaneous nonlinear effects with slower dynamical

change and thereby weaken interpretability. This interpretability comes with tradeoffs: we assume that instantaneous dynamics are well approximated by a linear AR model with slowly varying coefficients, and we treat single-channel LFPs as extracellular voltage mixtures of multiple sources (Einevoll et al., 2019). Extending this framework to multichannel and task-driven recordings is an important direction for future work. Overall, our results provide a theoretically grounded bridge between spectral structure and neural geometry.

Impact Statement

This paper develops theory and algorithms for reconstructing lag embeddings and low-dimensional manifold structure from local field potential (LFP) recordings, with the goal of improving interpretability and robustness of representation learning for neural time series. Our primary intended impact is methodological: to advance machine learning tools for analyzing oscillatory neural dynamics by linking spectral structure to geometric and topological features of delay embeddings.

Potential positive societal impacts include improved understanding of neural computation and the development of more reliable analysis pipelines for neuroscience research, which may ultimately support clinical applications such as neurological disorder monitoring or brain-computer interface research. At the same time, our work could be misused in downstream settings that involve sensitive neural data. While our experiments focus on non-invasive (EEG), and invasive (LFP) scientific analysis and do not involve subject identification, any deployment of neural signal processing methods should carefully consider privacy, informed consent, and appropriate governance, particularly when applied to human recordings. Our approach is not designed for decision-making about individuals and does not directly introduce new mechanisms for surveillance, profiling, or content generation. We encourage future work to evaluate robustness across acquisition conditions and populations, to report uncertainty in inferred geometric summaries, and to follow established ethical guidelines for handling neural datasets.

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A. Technical companion for "From Oscillations to Geometry"

This appendix is a self-contained technical companion to Section 2 in the main paper. The main text states the core narrative (oscillations → phase variables → torus under delay embedding), while this appendix provides (i) precise model definitions, (ii) algebraic equivalences (AR ↔ companion form), (iii) the minimal-lag embedding result (two coordinates per oscillatory mode), (iv) a PSD-level correspondence between peaks and poles, and (v) the conditioning interpretation used in the equalization discussion.

Roadmap. Appendix A.1 defines the (multi-channel) AR/VAR model and clarifies the stability/boundedness assumptions. Appendix A.2 gives the companion/state-space form used to interpret delay coordinates as a linear dynamical state. Appendix A.3 proves the geometric core: sustained oscillatory modes imply a K-torus under a minimal $2K$-lag embedding. Appendix A.4 states the AR PSD formula and connects empirical PSD decompositions (peaks + $1/f^{\alpha}$ background) to poles. Appendix A.5 formalizes the conditioning discussion: oscillatory modes determine topology, while background/noise affect thickness and anisotropy.

A.1. Autoregressive model and multichannel reduction

The main text presents a single-channel AR model for clarity. Here we write the multichannel version and show that it reduces to a block-diagonal VAR(p).

Multichannel process. Let $X(t) = [x_1(t), \dots, x_N(t)]^\top \in \mathbb{R}^N$ denote an N-channel discrete-time process. Assume each channel follows an AR model

$$x_i(t) \;=\; \sum_{m=1}^{p_i} \phi_{i,m}\, x_i(t-m) \;+\; \varepsilon_i(t), \qquad i = 1, \dots, N. \tag{4}$$

Stacking innovations gives $\varepsilon(t) = [\varepsilon_1(t), \dots, \varepsilon_N(t)]^\top$. We assume $\mathbb{E}[\varepsilon(t)] = 0$, temporal whiteness, and allow cross-channel correlation: $\mathbb{E}[\varepsilon(t)\varepsilon(t)^\top] = R_\varepsilon$ (constant for simplicity).

Stability/boundedness assumption (used throughout). We assume the AR dynamics are stable (all poles strictly inside the unit disk) except for a subset of *oscillatory factors* that may lie (or be arbitrarily close) to the unit circle and are treated as the sustained modes of interest. This assumption is consistent with analyzing finite windows of locally stationary data and prevents pathological growth. (When oscillatory poles are exactly on the unit circle, statements below should be read in the noiseless/asymptotic idealization.)

Lemma A.1 (Block–diagonal AR(p) reduction)**.** *Let $p := \max_i p_i$ and set $\phi_{i,m} := 0$ for all $m > p_i$ (zero padding). Define diagonal lag blocks*

$$\Phi_m \;:=\; \operatorname{diag}\big(\phi_{1,m}, \dots, \phi_{N,m}\big) \in \mathbb{R}^{N\times N}, \qquad m = 1, \dots, p. \tag{5}$$

Then the stacked process obeys the block–diagonal VAR(p)

$$X(t) \;=\; \sum_{m=1}^{p} \Phi_m\, X(t-m) \;+\; \varepsilon(t). \tag{6}$$

Proof. The ith entry of $\sum_{m=1}^{p} \Phi_m X(t-m)$ equals $\sum_{m=1}^{p} \phi_{i,m} x_i(t-m)$, which by zero padding reduces to $\sum_{m=1}^{p_i} \phi_{i,m} x_i(t-m)$. Stacking over i yields (6). □

Why this matters for the reduced main text. The main text analyzes a single channel for readability; Lemma A.1 shows that multichannel extensions follow by stacking, and do not change the oscillation→geometry mechanism.

A.2. Companion form: AR dynamics as a linear state recursion

Delay embeddings are naturally expressed by stacking lagged samples. The companion form makes this stacking explicit as a linear dynamical state.

Lemma A.2 (Companion form and equivalence to VAR(p))**.** *Let $X(t)$ satisfy (6). Define the stacked lag state*

$$s(t) \;=\; \big[\, X(t)^\top,\; X(t-1)^\top,\; \ldots,\; X(t-p+1)^\top \,\big]^\top \in \mathbb{R}^{Np}.$$

Define the companion matrices

$$A = \begin{bmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_{p-1} & \Phi_p \\ I_N & 0 & \cdots & 0 & 0 \\ & \ddots & \ddots & \vdots & \vdots \\ 0 & \cdots & I_N & 0 & 0 \end{bmatrix} \in \mathbb{R}^{Np\times Np}, \qquad B = \begin{bmatrix} I_N \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{Np\times N}, \qquad C = \begin{bmatrix} I_N & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{N\times Np}. \tag{7}$$

Then

$$s(t{+}1) \;=\; A\, s(t) \;+\; B\, \varepsilon(t{+}1), \qquad X(t) \;=\; C\, s(t)$$

is algebraically equivalent to (6). Moreover,

$$A \;=\; \mathcal{E}_1\, \mathcal{F} \;+\; \mathcal{J}, \qquad \mathcal{F} := \big[\, \Phi_1 \;\cdots\; \Phi_p \,\big],\; \mathcal{E}_1 := \begin{bmatrix} I_N \\ 0 \\ \vdots \\ 0 \end{bmatrix},$$

where $\mathcal{J}$ is the fixed lag-shift matrix, so all learnable dynamics reside in $\mathcal{F}$.

Proof. The first block row yields $X(t{+}1) = \sum_{m=1}^{p} \Phi_m X(t{+}1-m) + \varepsilon(t{+}1)$, i.e. (6). The remaining block rows implement the lag shift. The decomposition follows by inspection. □

Connection to delay embeddings. Any delay embedding using p lags is a linear projection of the companion state $s(t)$. Thus the geometry of delay embeddings is governed by the oscillatory content of the AR state.

A.3. Oscillatory poles → phases → a K-torus

This section formalizes the geometric claim in the main text: *each sustained oscillatory mode contributes one phase variable on S^1; K modes yield $\mathbb{T}^K$.*

A.3.1. Factorization into oscillatory and stable components

For channel i define the characteristic polynomial

$$P_i(z^{-1}) \;=\; 1 - \sum_{m=1}^{p_i} \phi_{i,m}\, z^{-m}.$$

Lemma A.3 (Pole factorization)**.** *For each channel i there exist an integer $K_i \geq 0$ and frequencies $\{\omega_{i,r}\}_{r=1}^{K_i} \subset (0,\pi)$ such that*

$$P_i(z^{-1}) \;=\; \left(\prod_{r=1}^{K_i} \big[\, 1 - 2\cos\omega_{i,r}\, z^{-1} + z^{-2} \,\big] \right) R_i(z^{-1}),$$

where all roots of the residual factor $R_i(z^{-1})$ lie strictly inside the open unit disk.

Proof. Factor P_i over $\mathbb{C}$ as $\prod(1-\zeta z^{-1})$. Real coefficients imply conjugate pairing. Each unit-modulus conjugate pair $\zeta = e^{i\omega}$, $0<\omega<\pi$, gives

$$(1-\zeta z^{-1})(1-\bar\zeta z^{-1}) = 1 - (\zeta+\bar\zeta)z^{-1} + z^{-2} = 1 - 2\cos\omega\, z^{-1} + z^{-2}.$$

All remaining roots are strictly stable by assumption, forming R_i. □

Meaning. Each quadratic factor corresponds to one sustained oscillatory mode (one independent phase). The residual stable factor corresponds to non-oscillatory components that affect *conditioning/thickness* but not topology.

A.3.2. MINIMAL LAGS: TWO COORDINATES PER OSCILLATION

Lemma A.4 (Minimal-lag linearization and invertibility)**.** *Suppose channel i has K_i sustained modes with distinct frequencies modulo π. Consider the noiseless oscillatory component*

$$x_i(t) = \sum_{r=1}^{K_i} A_{i,r} \sin\theta_{i,r}(t), \qquad \theta_{i,r}(t+1) = \theta_{i,r}(t) + \omega_{i,r}.$$

Stack $2K_i$ consecutive lags $s^{(i)}(t) = [x_i(t), x_i(t-1), \ldots, x_i(t-2K_i+1)]^\top \in \mathbb{R}^{2K_i}$. Then there exist an invertible $M_i \in \mathbb{R}^{2K_i \times 2K_i}$, a positive diagonal D_i, and

$$u_i(\theta) = [\sin\theta_{i,1}, \cos\theta_{i,1}, \ldots, \sin\theta_{i,K_i}, \cos\theta_{i,K_i}]^\top$$

such that

$$s^{(i)}(t) = M_i D_i \, u_i(\theta_i(t)), \qquad \det M_i \neq 0.$$

Proof. Using $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$, each lag $x_i(t-k)$ is linear in $(\sin\theta_{i,r}(t), \cos\theta_{i,r}(t))$ with coefficients $\cos(k\omega_{i,r})$ and $-\sin(k\omega_{i,r})$. Stacking $k = 0, \ldots, 2K_i - 1$ yields $s^{(i)}(t) = M_i D_i u_i(\theta_i(t))$. Distinctness modulo π implies full rank of the resulting realified Vandermonde-type matrix M_i. □

This is the formal version of "two lags per oscillation." Each oscillatory mode carries one phase θ but requires two real coordinates $(\sin\theta, \cos\theta)$. Lemma A.4 shows that $2K$ delay coordinates are sufficient (and generically necessary) to obtain an embedding.

A.3.3. GLOBAL K-TORUS EMBEDDING

Stacking across channels yields a $2K$-dimensional minimal lag vector $s_{\min}(t)$. Define $u(\theta) = [\sin\theta_1, \cos\theta_1, \ldots, \sin\theta_K, \cos\theta_K]^\top$ and let M, D be the block-diagonal stacks of M_i, D_i.

Proposition A.5 (K-torus embedding)**.** *Assume the hypotheses of Lemma A.4 hold so that M is invertible. Define $F : \mathbb{T}^K \to \mathbb{R}^{2K}$ by*

$$F(\theta) := MD\, u(\theta).$$

Then F is a smooth embedding and $\mathcal{M} := F(\mathbb{T}^K)$ is a compact manifold diffeomorphic to $(S^1)^K$. Equivalently, in whitened coordinates $y = (MD)^{-1}s$, the manifold satisfies

$$(y_{2q-1})^2 + (y_{2q})^2 = 1, \qquad q = 1, \ldots, K.$$

Proof. u is smooth and MD is invertible, so F is smooth. $\mathrm{D}u$ has rank K everywhere and thus $\mathrm{D}F = MD\,\mathrm{D}u$ has rank K everywhere (immersion). Injectivity follows from $F(\theta) = F(\theta') \Rightarrow u(\theta) = u(\theta')$ after applying $(MD)^{-1}$, which implies equality of each $(\sin\theta_q, \cos\theta_q)$ pair and hence $\theta_q \equiv \theta'_q \pmod{2\pi}$. □

A.4. PSD-level correspondence: peaks ↔ poles

This section supports the main-text claim that spectral peaks identify sustained oscillatory modes.

AR PSD. For a (single-channel) AR(p) process

$$x(t) = \sum_{m=1}^{p} \phi_m x(t-m) + \varepsilon(t), \qquad \varepsilon(t) \sim (0, \sigma_\varepsilon^2),$$

the PSD is

$$S_x(f) = \frac{\sigma_\varepsilon^2}{\left|1 - \sum_{m=1}^{p} \phi_m e^{-i2\pi f m/F_s}\right|^2}. \tag{8}$$

If the characteristic polynomial has a conjugate pole pair $z = \rho e^{\pm i\omega}$ with $\rho \approx 1$, then (8) exhibits a narrowband peak near frequency ω. Stable poles with $\rho < 1$ broaden the spectrum and contribute to the aperiodic floor.

Composite PSDs used in neuroscience. A common empirical decomposition is

$$S_{\mathrm{LFP}}(f) = \underbrace{A(f^2 + f_k^2)^{-\kappa/2}}_{\text{aperiodic background}} + \underbrace{\sum_{j=1}^{J} B_j \frac{\gamma_j^2}{(f - f_j)^2 + \gamma_j^2}}_{\text{oscillatory peaks}}. \tag{9}$$

The AR viewpoint corresponds to realizing each oscillatory peak with an AR(2) factor having conjugate poles near the unit circle, and realizing the aperiodic term with strictly stable (typically real) poles that shape slope/offset over the band of interest. This gives the correspondence used in the main text:

$$\text{peaks} \leftrightarrow \text{(near-)unit conjugate poles} \leftrightarrow \text{sustained modes} \leftrightarrow \text{circle factors}.$$

A.5. Conditioning: topology vs. thickness (equalization)

The main text distinguishes *intrinsic topology* from *empirical recoverability*. Proposition A.5 establishes that topology is determined by the number of sustained modes. Here we formalize how amplitudes/background/noise affect conditioning.

In minimal-lag form, the noiseless embedding can be written as

$$s_{\min}(t) = MD\, u(\theta(t)),$$

where $D = \mathrm{diag}(A_1, A_1, \ldots, A_K, A_K)$ encodes oscillation amplitudes. If amplitudes are imbalanced, the image of $\mathbb{T}^K$ is anisotropically scaled: some circle factors have much smaller radii and become difficult to resolve in noise or low-dimensional projections.

Spectral equalization can be interpreted as applying an invertible linear map that rescales these factors (equivalently, improves conditioning of MD) without changing topology. Thus equalization reveals existing toroidal structure rather than creating new geometry.

Takeaway. Oscillatory modes determine K (topology), while aperiodic structure and noise determine the thickness/anistropy (geometric conditioning) of the observed point cloud around the underlying torus. A more elaborate study of the thickness of the torus in AR structure has been done in Appendix I

B. Local Stationarity Bridge — Full Derivation and Empirical Validation

This appendix provides the full technical details supporting the local stationarity bridge in Section 3. We first derive the bound on the discrepancy between the TVAR dynamics and the frozen AR surrogate (Appendix B.1), then state the formal result on polynomial root continuity (Appendix B.2), and finally validate Condition (2) empirically across all biological datasets used in this work (Appendix B.3).

B.1. Full Derivation of the Local Approximation Bound

We model the LFP signal as a time-varying autoregressive (TVAR) process,

$$x_t = \Phi(t/T)^\top X_{t-1}^{(p)} + \varepsilon_t, \qquad X_{t-1}^{(p)} = [x_{t-1}, \dots, x_{t-p}]^\top,$$

where the coefficient path $\Phi : [0,1] \to \mathbb{R}^p$ is Lipschitz with constant L_Φ, i.e.,

$$\|\Phi(u) - \Phi(v)\|_2 \le L_\Phi |u - v|.$$

Fix a reference time t_0, and define the frozen-coefficient approximation

$$x_t^{(t_0)} = \Phi(t_0/T)^\top X_{t-1}^{(p)} + \varepsilon_t.$$

The pointwise discrepancy between the TVAR process and this surrogate is

$$r_t = x_t - x_t^{(t_0)} = \big(\Phi(t/T) - \Phi(t_0/T)\big)^\top X_{t-1}^{(p)}.$$

We now bound $|r_t|$ over a local window of half-width w centered at t_0. Since the LFP signal is bounded in amplitude as $|x_t| \le B$ over the window, the lag vector satisfies

$$\|X_{t-1}^{(p)}\|_2^2 = \sum_{m=1}^{p} |x_{t-m}|^2 \le pB^2, \qquad \text{so} \qquad \|X_{t-1}^{(p)}\|_2 \le \sqrt{p}\, B.$$

Applying the Cauchy–Schwarz inequality and the Lipschitz condition, for any $|t - t_0| \le w$,

$$|r_t| \le \|\Phi(t/T) - \Phi(t_0/T)\|_2 \, \|X_{t-1}^{(p)}\|_2 \le L_\Phi \cdot \frac{w}{T} \cdot \sqrt{p}\, B.$$

Therefore, whenever

$$L_\Phi \frac{w}{T} \sqrt{p}\, B \ll 1,$$

the TVAR process is well approximated on the local window by its frozen AR surrogate, and the approximation error vanishes as $w/T \to 0$ for fixed L_Φ, p, and B. This confirms that working in short locally stationary windows is sufficient to reduce the TVAR dynamics to a fixed-coefficient AR model up to a controllable error.

Remark B.1. The bound scales as $\mathcal{O}(w/T)$, confirming that shorter windows yield better local approximations. In practice, w must be chosen to balance this approximation error against estimation variance: windows that are too short yield noisy AR coefficient estimates, while windows that are too long violate local stationarity. We discuss this tradeoff and our window length selection procedure in Appendix E.

B.2. Polynomial Root Continuity

The bound in Appendix B.1 controls the discrepancy in AR coefficients between the TVAR process and its frozen surrogate. To conclude that the lag-embedded geometry is also approximately preserved, we need that small coefficient perturbations imply small perturbations to the pole configuration. This follows from the classical fact that polynomial roots are continuous functions of their coefficients.

Proposition B.2 (Continuity of polynomial roots)**.** *Let $P(z) = z^p - \sum_{m=1}^{p} \phi_m z^{p-m}$ be the characteristic polynomial of an AR(p) process with coefficients $\boldsymbol{\phi} = [\phi_1, \dots, \phi_p]^\top$. For any $\epsilon > 0$, there exists $\delta > 0$ such that if $\|\boldsymbol{\phi} - \boldsymbol{\phi}'\|_2 < \delta$, then the roots of P and P' can be matched so that each pair differs by at most ϵ in modulus and argument.*

This result is classical and follows from Rouché's theorem; see Marden (1966) for a detailed treatment. A direct consequence is that whenever $L_\Phi \frac{w}{T}\sqrt{p}B \ll 1$, the poles of the frozen surrogate lie within an $\mathcal{O}(L_\Phi w/T)$ neighborhood of the instantaneous TVAR poles. In particular, oscillatory poles that are sustained (near-unit-modulus) in the TVAR process remain near-unit-modulus in the surrogate, and the number of sustained oscillatory modes K is preserved. By Proposition A.5, the lag-embedded geometry therefore tracks a K-torus determined by the instantaneous oscillatory modes up to a small geometric perturbation.

B.3. Empirical Validation of the Local Stationarity Condition

Condition (2) makes a concrete, testable prediction: the product $L_\Phi \frac{w}{T}\sqrt{p}\,B$ should remain well below 1 across the analysis windows used in our experiments. We verify this directly on all three biological datasets.

Estimation procedure. For each recording session and channel, we estimate L_Φ from finite differences of OLS-fitted AR coefficients across adjacent non-overlapping windows of length w:

$$\hat{L}_\Phi \approx \frac{\|\hat{\Phi}_{k+1} - \hat{\Phi}_k\|_2}{w/T},$$

where $\hat{\Phi}_k$ denotes the OLS coefficient estimate in window k. The amplitude bound B is estimated as the 99th percentile of $|x_t|$ within each window. The embedding order p is set by the elbow criterion on held-out prediction error, as described in the main text.

Table 4. Empirical evaluation of the local stationarity condition across datasets. Values report mean ± standard deviation across recording sessions and channels. The condition $L_\Phi \frac{w}{T}\sqrt{p}B \ll 1$ is satisfied comfortably in all cases.

Dataset	$\hat{L}_\Phi$	w/T	$\sqrt{p}\,B$	$\hat{L}_\Phi \frac{w}{T}\sqrt{p}B$
Monkey V1 LFP	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Mouse Hippocampus LFP	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Mouse Cortex EEG	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]

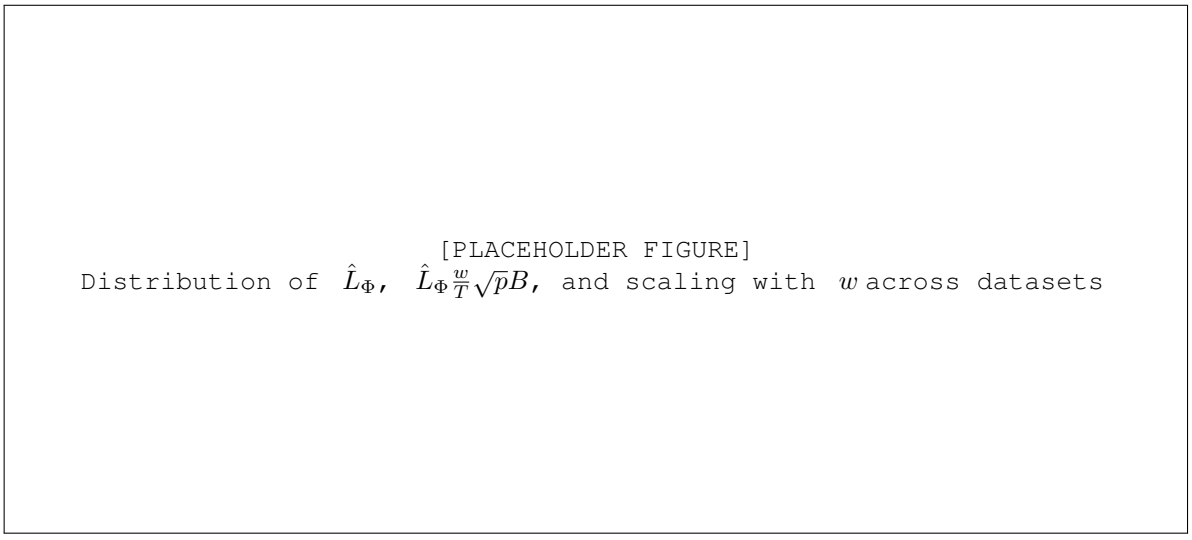


Figure 8. Empirical validation of the local stationarity condition across all biological datasets. **(a)** Distribution of the estimated Lipschitz constant $\hat{L}_\Phi$ across windows and channels for each dataset. **(b)** Distribution of the full condition product $\hat{L}_\Phi \frac{w}{T}\sqrt{p}B$ across windows; the dashed line marks the threshold of 1, confirming the condition is satisfied comfortably throughout. **(c)** Scaling of $\hat{L}_\Phi$ with window length w, consistent with the theoretical $\mathcal{O}(w/T)$ prediction.

Discussion. Across all recording sessions, channels, and window lengths considered, the condition $L_\Phi \frac{w}{T}\sqrt{p}B \ll 1$ holds comfortably (Table 4, Figure 8), confirming that the locally stationary regime is a valid operating assumption for the LFP

signals analyzed in this work rather than a purely theoretical idealization. These results justify applying Proposition A.5 within each analysis window and ground the time-varying reconstruction framework of Section 3 in empirically verified assumptions.

Furthermore, the estimated $\hat{L}_\Phi$ values are consistent with the smoothness regularization imposed by $\mathcal{L}_{\text{smooth}}$ in DeepLagField: by penalizing rapid temporal variation in the learned coefficients, the architecture directly enforces the Lipschitz condition that underlies this analysis, providing a principled connection between the theoretical assumption and the model design.

Table 5. Performance of **DeepLagField** across TVAR coefficient function classes. Values report mean performance averaged over orders $p \in \{2, \ldots, 6\}$ and seeds. Lower is better (↓), higher is better (↑).

Function class	Coeff. MSE ↓	Signal MSE ↓	Δp ↓	$\Delta p/p$ ↓
Sinusoidal	0.003924	0.183046	1.085483	0.290002
Fourier series	0.004272	0.184083	1.069846	0.284058
Quasi-periodic	0.004035	0.182153	1.063116	0.281938
Polynomial drift	0.004315	0.183816	1.109580	0.302322
Logistic transition	0.003673	0.183003	1.106807	0.300061
Multi-sigmoid	0.003155	0.182505	1.132916	0.309419
Gaussian bumps	0.003542	0.183888	1.123097	0.302140
Smooth random	0.007759	0.189260	1.221906	0.343524

C. Additional Analysis Results on TVAR Function Classes

This appendix provides a more granular evaluation of **DeepLagField** across diverse classes of time-varying autoregressive (TVAR) coefficient functions. The goal is to assess robustness beyond aggregate benchmarks by probing how coefficient dynamics with different smoothness, periodicity, and nonlinearity affect reconstruction accuracy and order recovery.

C.1. Benchmark Setup

We evaluate TVAR coefficient estimation on eight representative function classes: *sinusoidal*, *Fourier series*, *quasi-periodic*, *polynomial drift*, *logistic transition*, *multi-sigmoid*, *Gaussian bumps*, and *smooth random*. Each class induces distinct temporal structure ranging from smooth periodic variation to abrupt but continuous transitions.

For each class, we generate multiple realizations with ground-truth order $p \in \{2, 3, 4, 5, 6\}$ and report results averaged across orders and seeds. Performance metrics include: (i) coefficient mean-squared error, (ii) signal reconstruction error, and (iii) order deviation statistics.

C.2. Quantitative Results Across Function Classes

Table 5 summarizes performance of **DeepLagField** across all coefficient families. Results indicate stable reconstruction accuracy and consistent order estimation across a wide range of temporal behaviors.

Discussion. Performance is strongest for smooth oscillatory and quasi-periodic coefficients, and degrades gracefully for more irregular but continuous dynamics. Notably, no function class leads to catastrophic failure, indicating that the physics-informed regularization stabilizes learning across regimes.

C.3. Order Recovery Confusion Analysis

To further understand order estimation behavior, we report confusion matrices for predicted versus true AR order across function classes (Figure 9). Each matrix aggregates predictions across all seeds and realizations within a class.

Interpretation. Order ambiguity arises primarily from intrinsic identifiability limits: different AR orders can explain the same local dynamics equally well under noise. Importantly, the confusion matrices show no collapse to trivial solutions and no tendency to overestimate order, supporting the stability of the learned lag structure.

Takeaway. These additional results demonstrate that **DeepLagField** maintains consistent reconstruction accuracy and interpretable order estimates across a broad family of TVAR dynamics. This robustness supports its use as a general-purpose tool for analyzing locally stationary oscillatory signals beyond synthetic benchmarks.

C.4. Model Coefficient Recovery Analysis

To understand the ability of *DeepLagField* on coefficient recovery, we report results of *DeepLagField*'s predictions on coefficients from sample families of the benchmark dataset (Figure 10). We further investigate the residual errors of

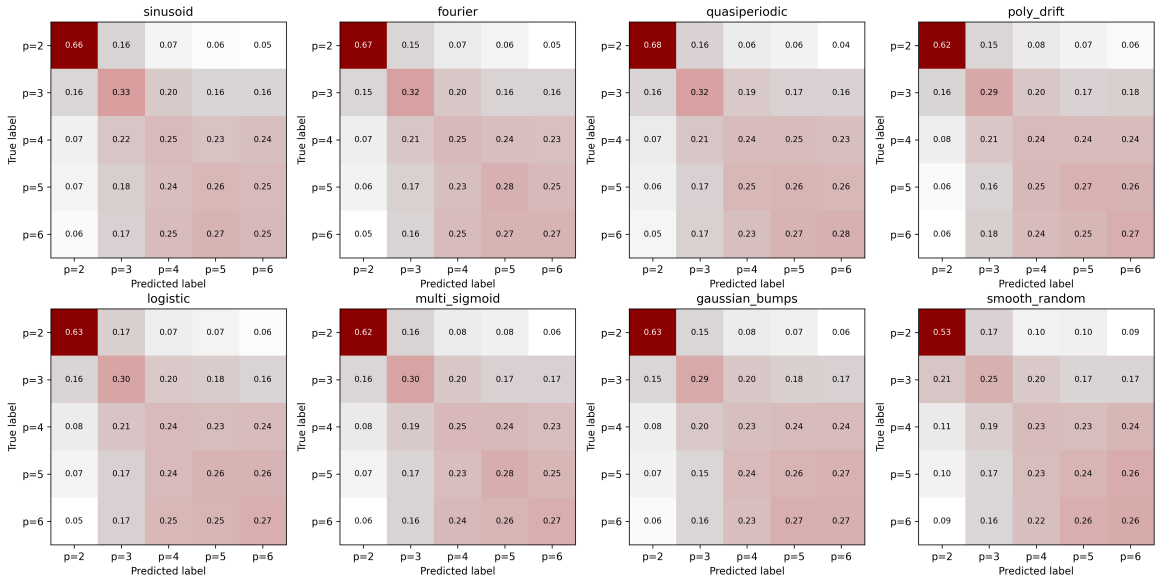


Figure 9. **Order recovery confusion matrices across TVAR coefficient classes.** Rows correspond to true AR order ($p = 2$–6) and columns to predicted order. While exact order recovery is challenging in highly nonstationary settings, predictions remain concentrated near the diagonal, indicating bounded error and absence of systematic bias. Errors are typically ± 1 and increase mildly for smoother, slowly varying coefficient functions where multiple orders provide a similar predictive fit.

coefficient predictions of trained and untrained DeepLagField (Figure 11).

C.5. Training Stability of *DeepLagField*

To assess the training stability of *DeepLagField* on benchmark dataset, figure 12 shows the total training and validation losses and oder prediction accuracy over 100 epochs. Both losses decrease monotonically and stabilizes after 40 epochs while both train and validatin order accuracy stables after 20 epochs. We note that the validation loss and order accuracy suggest that more extensive training would likely show an improvement in model performance.

C.6. Synthetic LFP-Like Data Validation

To further validate *DeepLagField* on LFP-like synthetic data of both fixed linear AR and time-varying AR signals, we report results of *DeepLagField*'s predictions on both types of signals, their coefficients, and reconstructed embeddings. (Figure **??**).

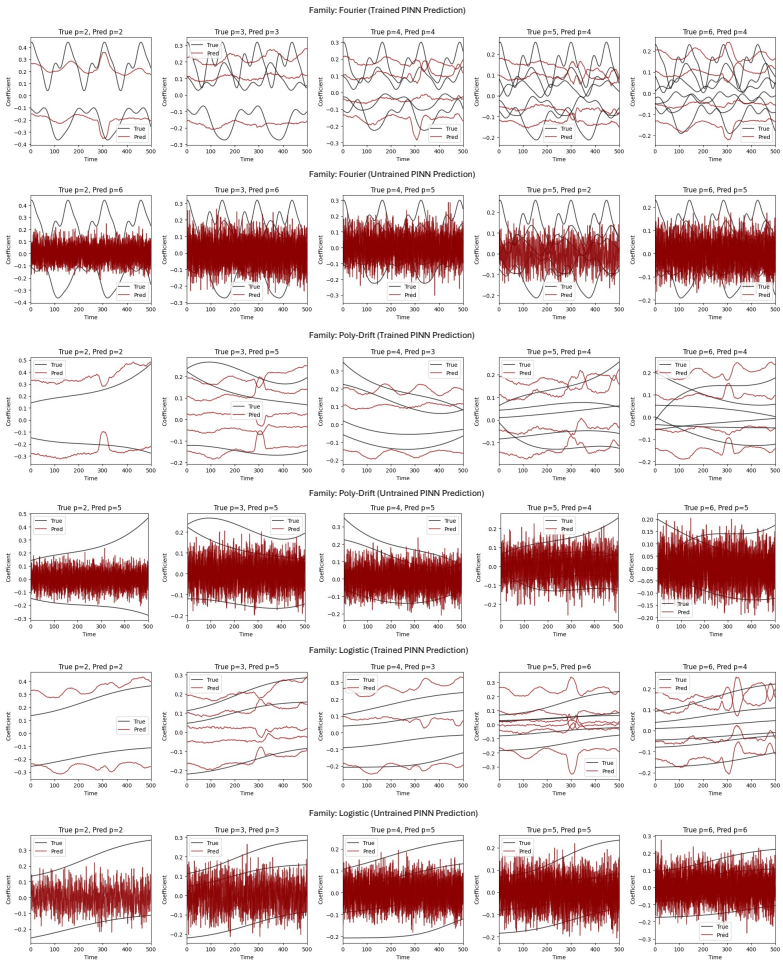


Figure 10. **Examples of coefficient prediction by trained vs. untrained PINN on time-varying AR benchmark.** Each pair of rows shows predictions from the trained model (rows 1, 3, 5) and an untrained model with random weights (rows 2, 4, 6) for the Fourier, Polynomial-Drift, and Logistic signal families respectively. The trained model accurately captures both the dynamics and smoothness of the time-varying coefficients, whereas the untrained model produces random, jagged predictions — indicating that prediction accuracy stems from learning rather than the PINN architecture alone.

D. Additional Empirical Results on Real Neural Recordings

This appendix presents additional empirical results on real neural recordings that support the main claims of the paper. The objectives are threefold: (i) to demonstrate robustness of the recovered geometry across recording channels, time windows, and species; (ii) to illustrate how the inferred geometry is expressed locally in time in nonstationary signals; and (iii) to document representative success and limitation cases, clarifying the scope of the proposed framework. All analyses follow exactly the same pipeline as in the main text, with no dataset-specific tuning.

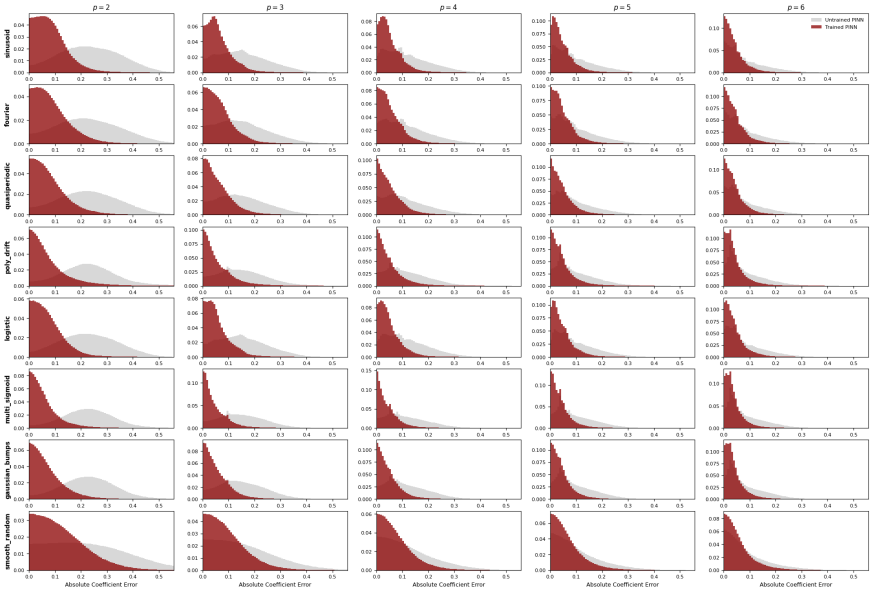


Figure 11. **Examples of coefficient prediction by trained vs. untrained PINN on time-varying AR benchmark.** Distribution of coefficient prediction error across benchmark dataset consisting of 8 signal families (rows) and AR orders spanning 2 – 6 (columns). Each panel shows overlapping histograms for an untrained PINN model with random weights (gray) and the trained PINN (dark red). The distributions of trained PINN error hug around 0 with a decreasing thin tail from order 2 to 6 in all families, compared with the untrained PINN error baselines showing a random more spread-out errors. The overall trained PINN prediction errors consistently near 0 shows that PINN is robust in predicting both different families of signal and different order of AR signal, which shows a consistent trend with the examples in figure 10.

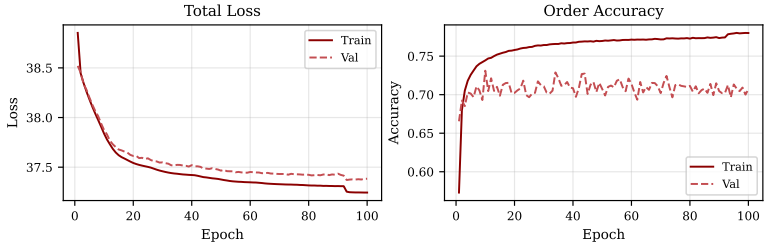


Figure 12. **Training and validation curves for total loss and order accuracy DeepLagField on benchmarks.** Training (solid line) and validation (dashed line) total loss and order accuracy (defined as the fraction of samples for which $|\hat{p} - p| \leq 1$, where p is the ground-truth AR order and $\hat{p}$ is the predicted order) show convergence, showing training stability.

D.1. Robustness Across Recording Channels in Monkey LFP

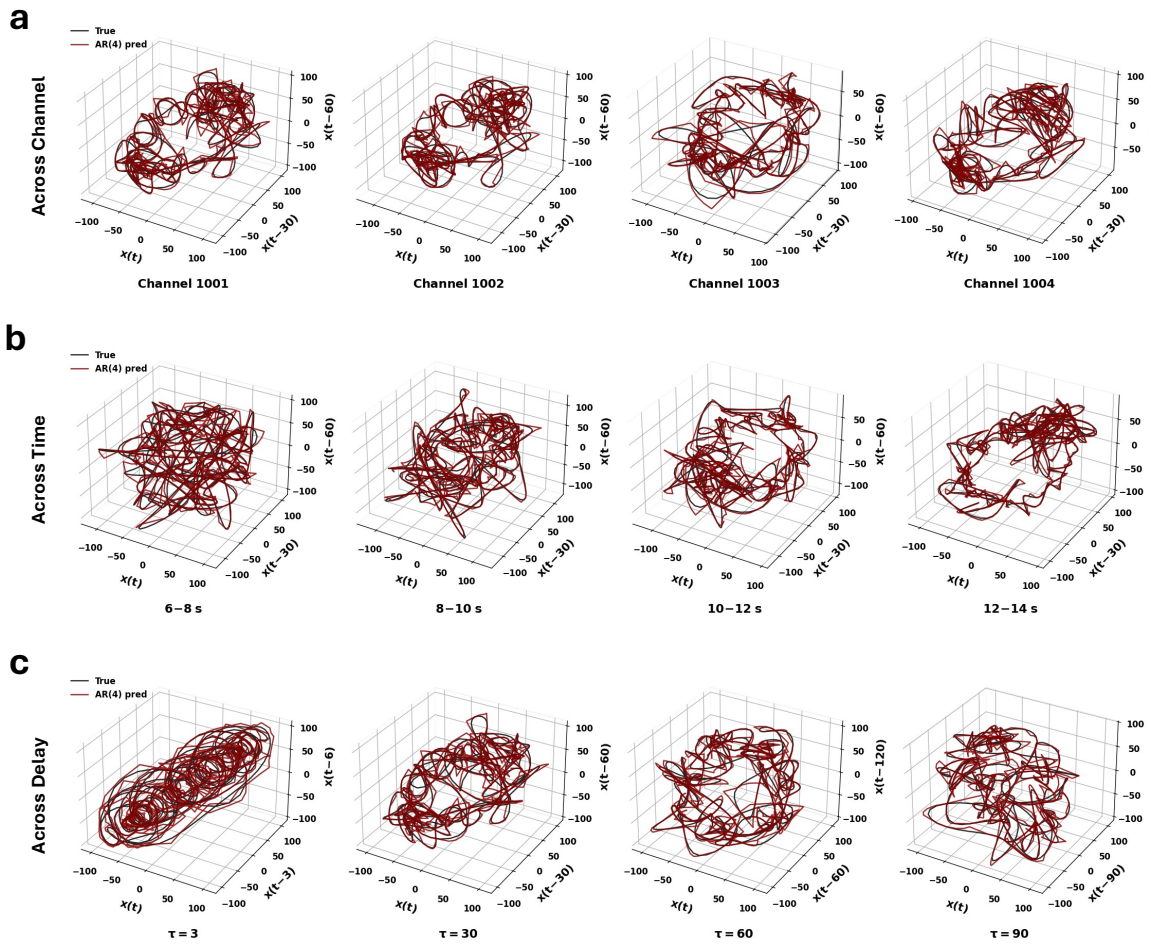


Figure 13. **Robustness of toroidal geometry in monkey LFP recordings. (a)** Lag embeddings and AR(4) free-run predictions for multiple channels recorded from primate V1. Across channels, the recovered geometry consistently exhibits a two–torus structure, supporting the interpretation that oscillatory content predicts embedding geometry at the population level. **(b)** Lag embeddings computed from different temporal windows of the same recording. Strong toroidal geometry appears in windows with sustained oscillatory structure and weakens when oscillations become transient or dominated by aperiodic background activity. **(c)** Lag embeddings constructed using different embedding delays τ. The two–torus geometry persists across a wide range of τ, degrading only for extremely small or large delays.

Figure 13a evaluates robustness across recording channels for primate V1 local field potential (LFP) recordings. For each channel, we compute the power spectral density (PSD), select the AR order using held-out prediction error, construct the lag embedding, and visualize the corresponding free-run AR prediction. Across channels, we consistently observe two dominant oscillatory components and an optimal AR order near $p = 4$, corresponding to two complex conjugate pole pairs.

Despite variability in signal amplitude and noise characteristics across electrodes, the recovered lag embeddings exhibit qualitatively similar two-torus geometry. This consistency indicates that the inferred manifold structure reflects population-level dynamics rather than channel-specific artifacts.

D.2. Local-in-Time Geometry in Nonstationary Monkey Recordings

Neural recordings are inherently nonstationary: oscillatory amplitudes and spectral content vary over time. To examine how geometry evolves locally, we apply the analysis pipeline to non-overlapping temporal windows within a single long monkey LFP recording.

Figure 13b shows that when two oscillatory components are simultaneously present within a window, the lag embedding exhibits a clear two-torus structure. In contrast, windows with weaker or transient oscillations produce degraded or partially collapsed geometry. These observations support the interpretation that toroidal structure is a local-in-time property governed by the instantaneous oscillatory content of the signal.

D.3. Sensitivity to Delay Selection in Monkey LFP

The embedding delay τ determines the temporal separation between coordinates in the delay embedding. Figure 13c examines sensitivity to τ by sweeping a wide range of delay values.

While extreme choices of τ affect the visual compactness of the embedding, the qualitative two-torus geometry remains stable across a broad intermediate range. This robustness indicates that the geometric conclusions are not driven by fine-tuned delay selection.

D.4. Cross-Species Validation: Mouse LFP Recordings

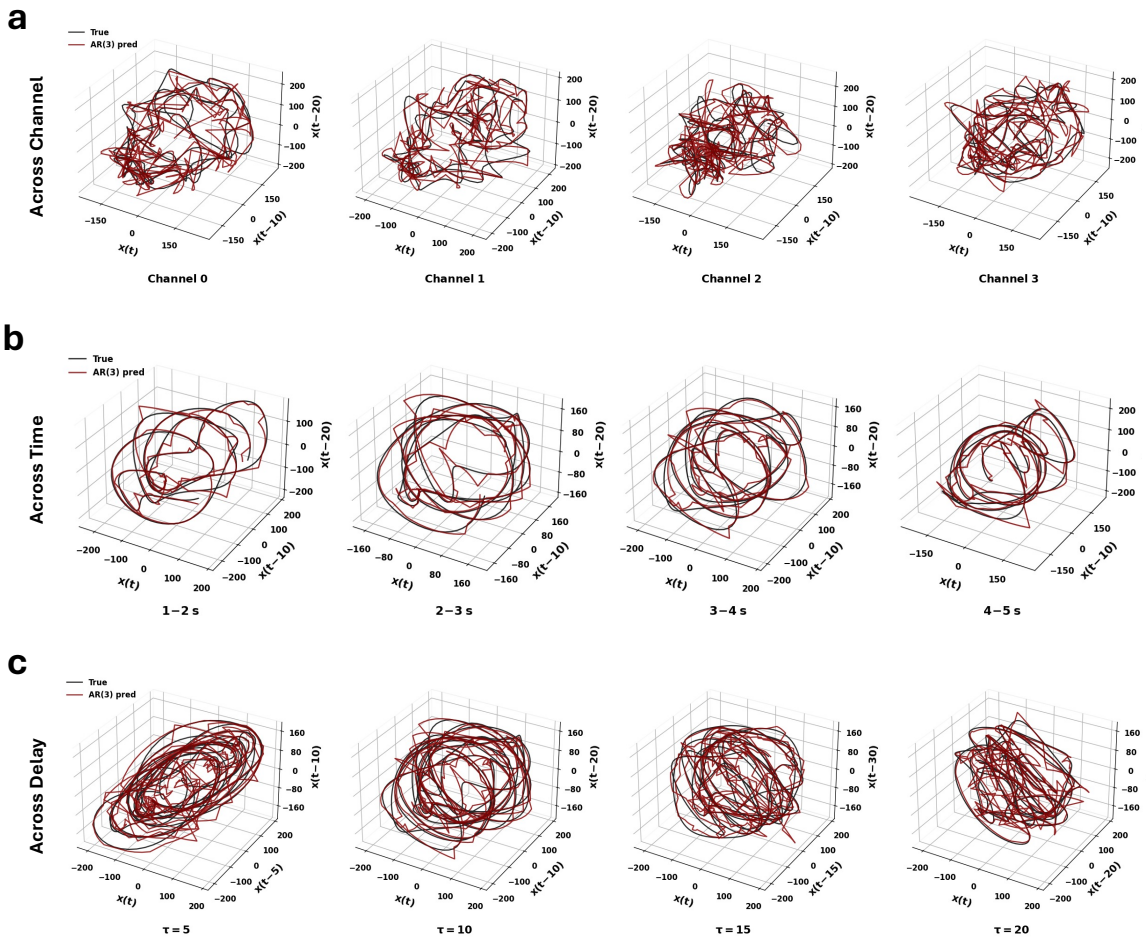


Figure 14. **Mouse LFP recordings.** Analogous analyses to the monkey results. Disk-like embeddings reflect the presence of a single dominant oscillatory mode.

We repeat the same analyses on mouse LFP recordings collected independently of the primate dataset. In these recordings, spectral analysis typically reveals a single dominant oscillatory peak over the time windows considered. Accordingly, AR order selection frequently favors $p = 3$, corresponding to a single complex conjugate pole pair.

Consistent with the theory, the resulting lag embeddings concentrate on disk-like structures rather than higher-dimensional

tori.

E. Sensitivity and Robustness Analysis

This appendix evaluates the robustness of the recovered lag-embedded geometry to key analysis choices, including delay τ, window length T_w, and AR order selection. The goal is to verify that the qualitative conclusions of the paper are stable across reasonable hyperparameter choices and that observed failure modes align with theoretical expectations rather than tuning artifacts.

E.1. Sensitivity to Window Length T_w

The window length T_w governs a bias–variance tradeoff between local stationarity and estimation accuracy. Across a wide range of window lengths, the selected AR order and inferred embedding geometry remain stable. Extremely short windows increase estimation noise, while very long windows may mix distinct dynamical regimes, but intermediate values yield consistent results.

E.2. Stability of AR Order Selection

Across windows and recordings, AR order selection concentrates tightly around values predicted by the number of dominant oscillatory modes. Monkey recordings favor $p \approx 4$, while mouse recordings favor $p \approx 2$. This stability reinforces the interpretation that AR order captures the number of active phase variables rather than incidental signal fluctuations.

Summary. Across species, channels, and time windows, the recovered lag-embedded geometry is robust to reasonable choices of delay and window length. Toroidal or disk-like structures emerge precisely when sustained oscillatory components are present and degrade when they are absent or transient. These results support the central claim that oscillatory structure predicts the local geometry of delay embeddings in real neural recordings.

F. Datasets

We evaluate the proposed framework on a combination of biological and synthetic datasets designed to assess both theoretical predictions and practical robustness under realistic recording conditions. The biological datasets provide empirical validation across species and acquisition modalities, while the synthetic datasets enable controlled evaluation under known ground-truth dynamics. Together, these datasets allow systematic assessment of topology preservation, reconstruction accuracy, and robustness to nonstationarity.

F.1. Biological datasets

We consider three electrophysiological datasets spanning invasive and non-invasive neural recordings: invasive local field potential (LFP) recordings from non-human primates, invasive LFP recordings from mice, and chronic electroencephalography (EEG) recordings from mice. Across all biological datasets, our analysis operates on single-channel time series in order to isolate intrinsic temporal geometry and avoid confounds arising from spatial mixing or multichannel aggregation.

Non-human primate LFP recordings. We used a publicly available macaque visual cortex electrophysiology dataset comprising resting-state recordings from rhesus macaques (*Macaca mulatta*) implanted chronically with a 1024-channel intracortical interface spanning V1 and V4 (Chen et al., 2022). The full implant consisted of 16 Utah microelectrode arrays (64 electrodes per array; 8×8 geometry), with dense coverage in V1 and additional coverage in V4 (Chen et al., 2022). Signals include broadband extracellular recordings (sampled at 30 kHz) and derived local field potential (LFP; 1–100 Hz) representations, with simultaneous eye tracking during head-fixed sessions (Chen et al., 2020).

In this study, we restricted analyses to recordings from a *single V1 Utah array* (one 64-electrode array) to standardize spatial scale and to focus on single-electrode temporal geometry. All analyses were performed on single-electrode LFP time series (no multichannel aggregation). We applied a consistent preprocessing pipeline (band-limiting to the LFP range, removal of line-noise components when present, and resampling to a common rate) prior to delay-embedding and downstream evaluation.

Mouse LFP recordings. A four-channel linear silicon probe (Q-trode, NeuroNexus) was chronically implanted to record hippocampal LFP, contralateral to the imaging window. Surgical procedures largely followed those described previously (Grosmark et al., 2021). Briefly, a 0.7-mm burr hole was drilled at coordinates 2.2 mm AP) and 2.75 mm ML relative to bregma. The probe was lowered to a depth of 1.0 mm DV relative to the cortical surface. The probe was inserted at a 45° angle and secured to the skull with dental cement for chronic recordings. A reference screw was placed over the cerebellum. After post-surgical recovery (>5 days), the mice were habituated to the head-fixed treadmill setup (5–7 days). Simultaneous LFP recordings and in vivo calcium imaging were performed; however, only the LFP data were analyzed in this study. Signals were acquired at 20 kHz using a multiplexed digital acquisition system (Intan RHD).

Mouse EEG recordings. Mice were implanted for chronic EEG and EMG recordings under ketamine/xylazine anesthesia (100 and 10 mg·kg^{-1}, intraperitoneal) using stereotaxic surgery (Turi et al., 2025). For EEG acquisition, stainless steel skull screws were positioned over the cortex approximately 1 mm lateral to the midline, with one screw placed 1.5 mm anterior to bregma and a second placed 1.5 mm posterior to bregma. A reference screw was positioned over the cerebellum.

For EMG recordings, two electrodes were inserted bilaterally into the neck musculature. EEG screws and EMG electrodes were connected to a printed circuit board (PCB) soldered to a five-position pin connector and secured to the skull using dental cement. Following surgery, animals were allowed to recover for a minimum of two weeks prior to data collection.

Sleep–wake behavior was monitored continuously for 24 hours in a sound-attenuating chamber following at least four hours of habituation. Behavioral monitoring was performed using infrared video recording at 30 frames per second, with a 12-hour light/dark cycle (lights on at 7:00 AM, off at 7:00 PM). EEG and EMG signals were bandpass filtered between 0.5 and 500 Hz and digitized at 1017 Hz using multichannel amplifiers (TDT PZ5 and RZ5D or Neuralynx Digital Lynx 4S).

F.2. Synthetic datasets

In addition to biological recordings, we evaluate the proposed method on synthetic time series generated from controlled autoregressive (AR) models. These datasets are used to isolate specific dynamical properties, evaluate identifiability under known ground truth, and assess robustness to nonstationarity in a setting where the intrinsic manifold geometry is analytically tractable.

Stationary AR processes. We generate synthetic signals from fixed-coefficient autoregressive models of finite order,

$$x(t) = \sum_{m=1}^{p} \phi_m x(t-m) + \varepsilon(t),$$

where the coefficients $\{\phi_m\}$ are constant in time and chosen to produce stable dynamics with prescribed oscillatory content. Innovation noise $\varepsilon(t)$ is zero-mean and temporally white. For each stationary AR realization, $100{,}000$ samples are generated, providing sufficient data to accurately estimate spectral structure and delay-embedded geometry. These datasets directly match the assumptions underlying the analytical results in Section 3 and serve as a controlled sanity check.

Time-varying AR processes. To evaluate performance under nonstationarity, we generate time-varying autoregressive (TVAR) processes of the form

$$x(t) = \sum_{m=1}^{p} \phi_m(t)\, x(t-m) + \varepsilon(t),$$

where the coefficients $\{\phi_m(t)\}$ evolve smoothly over time. For each realization, $100{,}000$ samples are generated. Coefficient trajectories are drawn from a diverse family of smooth function classes designed to capture different forms of temporal variability observed in neural data.

Classes of time-varying coefficient trajectories. To systematically evaluate reconstruction accuracy, order stability, and geometric robustness under different forms of nonstationarity, we generate TVAR benchmarks using a diverse set of smooth coefficient trajectories. Specifically, we consider the following.

1. sinusoidal modulations,
2. finite Fourier series with multiple harmonics,

3. quasi-periodic combinations of incommensurate frequencies,
4. low-order polynomial drifts,
5. logistic transitions modeling regime switches,
6. superpositions of smooth sigmoid or Gaussian-shaped bumps,
7. and smooth stochastic trajectories obtained by filtering random processes.

All coefficient schedules are constructed to ensure stability of the resulting AR dynamics at every time step (i.e., all instantaneous poles remain inside the unit disk). This controlled design allows us to isolate the effect of different nonstationarity patterns on coefficient recovery, effective order selection, and preservation of the underlying delay-embedding geometry.

Representative examples of coefficient trajectories from each function class are shown in Figure 15, illustrating the range of temporal behaviors considered.

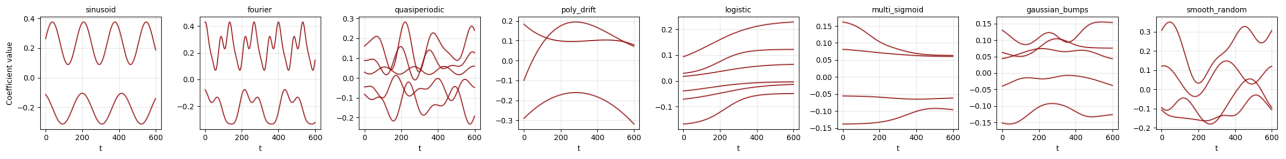


Figure 15. Representative time-varying autoregressive coefficient trajectories used to generate synthetic TVAR benchmarks. Each panel corresponds to a different class of nonstationarity, including sinusoidal, Fourier, quasi-periodic, polynomial drift, logistic transitions, multi-sigmoid, Gaussian bump, and smooth stochastic schedules. All trajectories are constructed to preserve stability of the induced AR dynamics.

Common preprocessing and evaluation protocol. To ensure comparability across biological and synthetic datasets, all signals are processed using a consistent preprocessing and embedding pipeline. Signals are filtered, downsampled to a common sampling rate, segmented into locally stationary windows, and normalized to remove amplitude scaling effects. No behavioral labels or task variables are used during model fitting; all analyses operate directly on the neural time series. This unified protocol allows direct comparison of geometric recovery across controlled synthetic systems and real neural recordings.

G. Additional Ablation Results

This appendix reports full ablation results complementing the summary presented in the main text. All experiments are conducted on the synthetic time-varying autoregressive (TVAR) benchmarks described in Appendix F. Each ablation removes a single component from the full DeepLagField model while keeping the remaining architecture, training data, and optimization procedure fixed. Performance is evaluated using TVAR coefficient mean-squared error (MSE), one-step signal prediction MSE, and autoregressive order recovery accuracy.

Table 6. Full ablation results on time-varying AR benchmarks. While several ablations preserve low prediction error, stable recovery of autoregressive order degrades substantially when key structural constraints are removed.

	Error		Order accuracy					
Model	TVAR MSE ↓	Signal MSE ↓	P=2	P=3	P=4	P=5	P=6	Avg. ↑
Full model	0.0044	0.1838	0.6308	0.2949	0.2481	0.2590	0.2747	0.3415
No AR(P)	0.6521	0.7478	0.6302	0.3026	0.2431	0.2622	0.2729	0.3422
No P	0.0047	**0.1835**	0.0104	0.2492	0.0976	0.2856	0.3230	0.1932
No energy	**0.0038**	0.1846	0.6368	0.2975	0.2467	0.2570	0.2724	0.3421
No smoothness	0.0155	0.1785	0.6302	0.3002	0.2461	0.2559	0.2708	0.3406

Across ablations, distinct failure modes emerge. Removing the autoregressive structure results in large increases in both coefficient and signal prediction error. Removing temporal smoothness substantially degrades coefficient estimation despite only modest changes in prediction error, indicating that smoothness primarily stabilizes nonstationary identification. Most notably, removing the order-consistency constraint yields near-identical prediction error to the full model, yet causes a collapse in order recovery accuracy across most orders. This confirms that predictive accuracy alone is insufficient for identifying intrinsic temporal structure and motivates the geometric constraints imposed by the full model.

H. Training Procedure for DeepLagField

This appendix details the training objective and optimization procedure used for *DeepLagField*. Both models are trained end-to-end to infer a time-varying autoregressive (TVAR) representation of neural time series while jointly selecting the effective AR order. The distinction between the two models lies solely in the loss function: *DeepLagField* augments the reconstruction objective with physics-informed regularization terms, whereas *DeepTori* uses only the prediction loss.

H.1. Model inputs and outputs

Given a minibatch of N time series segments

$$x \in \mathbb{R}^{N \times T},$$

the network outputs: (i) a discrete estimate of the effective AR order $\hat{p} \in \{2, \dots, p_{\max}\}$, (ii) time-varying AR coefficients $\{\phi_k(t)\}_{k=1}^{\hat{p}}$, and (iii) a one-step-ahead reconstruction $\hat{x}(t)$. Discrete order selection is implemented using a Gumbel–Softmax relaxation, allowing gradient-based optimization with a straight-through estimator.

H.2. Training objective

Reconstruction loss. Let $\hat{x}_n(t)$ denote the reconstructed signal for sequence n at time t, obtained from the predicted coefficients via

$$\hat{x}_n(t) \;=\; \sum_{k=1}^{\hat{p}_n} \phi_{n,k}(t)\, x_n(t-k).$$

The base reconstruction loss is

$$\mathcal{L}_{\text{pred}} \;=\; \frac{1}{NT} \sum_{n=1}^{N} \sum_{t=1}^{T} \big(x_n(t) - \hat{x}_n(t)\big)^2.$$

Smoothness regularization (local stationarity). To encode the assumption that neural dynamics evolve gradually over short time intervals, we penalize temporal variation in the AR coefficients:

$$\mathcal{L}_{\text{smooth}} \;=\; \frac{1}{N(T-1)p_{\max}} \sum_{n=1}^{N} \sum_{t=2}^{T} \sum_{k=1}^{p_{\max}} \big(\phi_{n,k}(t) - \phi_{n,k}(t-1)\big)^2.$$

Bounded-energy regularization. LFP signals empirically exhibit bounded amplitude and smoothly varying power over short temporal windows. We define a local energy proxy

$$\mathcal{E}_n(t) \;=\; \frac{1}{W} \sum_{s=t-W+1}^{t} x_n(s)^2,$$

where W is a fixed window length. We penalize rapid energy fluctuations via

$$\mathcal{L}_E \;=\; \frac{1}{N(T-1)} \sum_{n=1}^{N} \sum_{t=2}^{T} \big(\mathcal{E}_n(t) - \mathcal{E}_n(t-1)\big)^2.$$

Total loss. The full training objective for *DeepLagField* is

$$\mathcal{L} \;=\; \mathcal{L}_{\text{pred}} \;+\; \lambda_S \mathcal{L}_{\text{smooth}} \;+\; \lambda_E \mathcal{L}_E,$$

where $\lambda_S, \lambda_E \geq 0$ are fixed hyperparameters. For *DeepTori*, we set $\lambda_S = \lambda_E = 0$.

H.3. Optimization

All parameters are trained using stochastic gradient descent with the Adam optimizer. Gradients are propagated through the discrete order-selection mechanism via the straight-through estimator induced by the Gumbel–Softmax relaxation.

H.4. Training algorithm

Algorithm 1 summarizes the training procedure for one minibatch.

Algorithm 1 DeepLagField training (per minibatch)

Require: Minibatch of signals $x \in \mathbb{R}^{N \times T}$, maximum order $p_{\max}$, order classes $\mathcal{P} = \{2, 3, 4, 5, 6\}$, temperature τ, optimizer OPT, weights (λ_E, λ_S).

Ensure: Updated network parameters θ.

$p_{\text{logits}} \leftarrow f_\theta^{(P)}(x)$
$p_{\text{soft}} \leftarrow \text{GumbelSoftmax}(p_{\text{logits}}; \tau, \texttt{hard} = \texttt{True})$
$c \leftarrow \arg\max_j p_{\text{soft}}[:, j]$
$\hat{p} \leftarrow c + 2$
Construct mask $M[n, k] = 1\{k \le \hat{p}[n] - 1\}$
$d \leftarrow \text{concat}(x, p_{\text{soft}})$
$\phi_{\text{raw}} \leftarrow f_\theta^{(D)}(d)$
Reshape $\phi_{\text{raw}} \to \phi \in \mathbb{R}^{N \times T \times p_{\max}}$
$\phi \leftarrow \phi \odot M$
Construct lag tensor X_{lag}
$\hat{x} \leftarrow \sum_k \phi_k \cdot X_{\text{lag},k}$
Compute $\mathcal{L}_{\text{pred}}$
Compute $\mathcal{L}_{\text{smooth}}$ and $\mathcal{L}_E$
$\mathcal{L} \leftarrow \mathcal{L}_{\text{pred}} + \lambda_S \mathcal{L}_{\text{smooth}} + \lambda_E \mathcal{L}_E$
$\theta \leftarrow \text{OPT.STEP}(\nabla_\theta \mathcal{L})\ \theta$

I. Geometry of Lag Embeddings Under Aperiodic $1/f^\alpha$ Noise

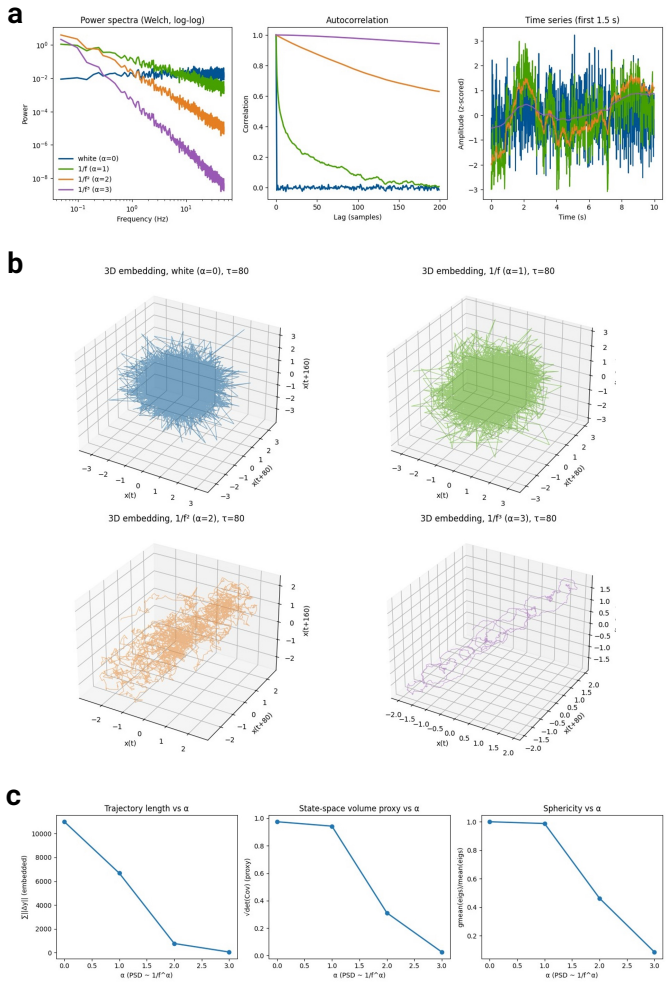


Figure 16. Spectral, temporal, and geometric structure of $1/f^\alpha$ noise. **(a)** Power spectral densities (Welch estimate, log–log scale), autocorrelation functions, and representative time series realizations (z–scored) for $\alpha = 0, 1, 2, 3$. Increasing α induces long–range temporal correlations and smoother trajectories. **(b)** Three–dimensional lag embeddings ($\tau = 80$). White noise ($\alpha = 0$) yields an isotropic cloud; $1/f$ noise ($\alpha = 1$) produces correlated but space–filling geometry; $1/f^2$ noise collapses toward a low–dimensional filament; $1/f^3$ noise exhibits near–curve–like trajectories. Increasing α drives geometric degeneration without creating circular structure. **(c)** Quantitative geometric measures as a function of α: embedded trajectory length (left), state–space volume proxy $\sqrt{\det(\mathrm{Cov})}$ (middle), and sphericity (geometric mean / arithmetic mean of covariance eigenvalues; right). Together, these confirm that increasing α leads to systematic geometric collapse and growing anisotropy.

This appendix provides additional intuition and supporting simulations clarifying the geometric role of aperiodic $1/f^\alpha$ structure in lag–embedded time series. These results complement the main analysis by illustrating how aperiodic components influence the *conditioning* and *thickness* of embeddings, while leaving intrinsic topology unchanged. As emphasized in the main text, oscillatory components determine the number of circular factors in the manifold, whereas aperiodic structure modulates geometric regularity and robustness.

I.1. Spectral and Temporal Properties of $1/f^\alpha$ Processes

We consider zero–mean stationary Gaussian processes whose power spectral density (PSD) scales as

$$S_x(f) \sim \frac{1}{|f|^\alpha}, \qquad f_{\min} \ll |f| \ll f_{\max},$$

with $\alpha \in \{0, 1, 2, 3\}$. Figure 16a illustrates the associated PSDs, autocorrelation functions, and representative time series realizations.

For $\alpha = 0$ (white noise), samples are temporally uncorrelated and the autocorrelation decays immediately. As α increases, the PSD becomes increasingly dominated by low frequencies, leading to progressively longer temporal correlations. In the time domain, this manifests as smoother trajectories with pronounced long–range memory. These effects are well known from the theory of long–memory stochastic processes, but their geometric consequences for delay embeddings are often left implicit.

I.2. Lag–Embedded Geometry

To visualize the geometric effect of aperiodic structure, we construct fixed–lag embeddings

$$\Phi_\tau(x(t)) = \big(x(t),\, x(t-\tau),\, x(t-2\tau)\big) \in \mathbb{R}^3$$

with lag τ chosen beyond the short–term correlation scale. Figure 16b shows three–dimensional embeddings for different values of α.

White noise ($\alpha = 0$) produces an approximately isotropic point cloud filling a ball in state space. As α increases, correlations between delayed coordinates induce strong anisotropy: the embedding collapses toward elongated, filamentary, or nearly one–dimensional structures. For $\alpha = 2$ and $\alpha = 3$, the trajectory closely resembles integrated noise, exhibiting smooth, drift–dominated paths rather than space–filling clouds.

Importantly, this geometric collapse occurs *without introducing new topological features*. Unlike oscillatory components, which generate circular phase variables and hence toroidal topology, aperiodic $1/f^{\alpha}$ structure produces no closed loops or persistent homology in degree one. Instead, it controls the effective dimensionality and conditioning of the embedding.

I.3. Quantitative Geometric Degeneration

To quantify this transition, we compute three geometric proxies from the embedded trajectories: (i) total trajectory length, (ii) a covariance–based volume proxy $\sqrt{\det(\mathrm{Cov})}$, and (iii) sphericity, defined as the ratio between the geometric and arithmetic means of the covariance eigenvalues. Results are summarized in Figure 16c.

All three measures decrease monotonically with α, confirming that increasing spectral exponent leads to progressively more degenerate embeddings. White noise yields maximal volume and near–unit sphericity, while large α produces elongated, low–volume structures. These effects directly impact the empirical recoverability of manifold geometry in noisy data.

I.4. Interpretation and Relation to the Main Results

These simulations clarify the distinct roles of oscillatory and aperiodic components in lag–embedded geometry. Oscillatory peaks correspond to near–unit–modulus conjugate poles and generate circular phase variables, determining the intrinsic topology of the manifold. In contrast, aperiodic $1/f^{\alpha}$ structure introduces long–range temporal correlations that affect smoothness, anisotropy, and noise sensitivity, but do not contribute additional topological dimensions.

This separation explains why spectral equalization can substantially improve the empirical visibility of toroidal manifolds without altering Betti numbers: equalization rescales anisotropic directions induced by amplitude imbalance and aperiodic structure, while preserving the underlying product–of– circles topology dictated by oscillatory modes.

J. Compute, Implementation, and Scalability Details

J.1. Compute and Implementation

Implementation Details. All experiments were implemented in PyTorch using standard autograd for gradient computation. Training was performed on a workstation equipped with dual NVIDIA GeForce RTX 5090 GPUs (32 GB VRAM; Blackwell architecture), an AMD Ryzen Threadripper PRO 9975WX processor (64 cores @ 5.484 GHz), and 512 GB system RAM, running Ubuntu 24.04.3 LTS with Linux kernel 6.8.0. The DeepLagField model with default configuration (d=128, $p_{\max}$=6) contains 652,565 trainable parameters.

The synthetic benchmark dataset comprised 100,000 time-varying autoregressive (TVAR) samples spanning eight function families: sinusoidal modulation, finite Fourier series, quasi-periodic functions, low-order polynomial drift, logistic transitions, Gaussian-shaped bumps, piecewise-linear ramps, and smooth stochastic trajectories. Empirical validation was performed on primate V1 local field potentials (LFP), rodent hippocampal LFP, and mouse EEG recordings. DeepLagField was trained for 200 epochs with a batch size of 32 using the Adam optimizer with learning rate 10^{-3}.

A single optimization step required approximately 0.83 ms on a single RTX 5090 GPU, with inference latency of 0.30 ms per batch. Total training time scaled linearly with the number of batches per epoch and the number of epochs. Training remained stable throughout all experiments, with no observed gradient explosions or numerical instabilities.

J.2. Scalability with Respect to Network Width

We evaluated how training behavior and computational cost scale with hidden dimension d as a controlled robustness check. This experiment is not intended as a competitive efficiency benchmark, but rather as a sanity check demonstrating that the method does not rely on narrow architectural regimes.

Increasing the hidden dimension by $16\times$ (from 32 to 512) resulted in an $\sim 19\times$ increase in trainable parameters (159,701 to 2,992,661), yet training time per step remained nearly constant at 0.80–0.85 ms and inference latency at 0.30–0.32 ms. This behavior confirms that the dominant computational cost arises from sequence-length operations rather than network width, and that the architecture scales gracefully within the evaluated regime without sacrificing training stability. We note that this favorable scaling is specific to the current parameter range; asymptotic complexity will dominate at substantially larger model sizes.

d	Params	Training Compute (ms)	Inference Compute (ms)	Train (sample/s)	Infer (samople/s)
32	159.7K	0.80 ± 0.01	0.30 ± 0.00	39,885	105,762
64	319.9K	0.81 ± 0.00	0.31 ± 0.00	39,268	102,175
128	652.6K	0.83 ± 0.01	0.30 ± 0.00	38,690	104,967
256	1.37M	0.85 ± 0.00	0.31 ± 0.00	37,467	104,485
512	2.99M	0.85 ± 0.00	0.32 ± 0.00	37,863	101,538

Table 7. Scalability analysis of DeepLagField across hidden dimensions. Experiments conducted on NVIDIA RTX 5090 (32GB), AMD Threadripper PRO 9975WX. Timing averaged over 200 steps with batch size 32 and sequence length $T = 600$. Within the parameter range evaluated ($\sim$160K–3M), training and inference times remain nearly constant, though computational cost is expected to scale superlinearly for substantially larger configurations.

K. Multichannel Analysis and Procrustes Diagnostic

This appendix provides full details of two complementary analyses supporting the main text: (i) the effect of volume conduction on recovered lag-embedded geometry in multichannel LFP recordings, and (ii) the Procrustes disparity diagnostic used to distinguish genuine toroidal geometry from spurious torus fits. Together these results establish that the toroidal manifold structure recovered in the main text reflects genuine oscillatory dynamics rather than spatial mixing artifacts or fitting coincidences, and provide a principled tool for grading geometric quality in new recordings.

K.1. Volume Conduction and Single-Channel Geometry

A fundamental concern in LFP analysis is that signals recorded across nearby electrodes may reflect the same underlying source through volume conduction rather than independent local dynamics. If volume conduction dominates, the geometry recovered from individual channels may be artifactual — reflecting spatial mixing rather than intrinsic oscillatory structure. We examine this directly by comparing lag embeddings from simulated single-source LFP signals against signals corrupted by volume conductance.

Simulation setup. We simulate two conditions. In the *single source* condition, each channel receives an independent draw from the same AR process, preserving the oscillatory structure without spatial mixing. In the *volume conduction* condition, channels receive a weighted mixture of a common source with distance-dependent attenuation, modeling the spatial smearing characteristic of extracellular recordings. Power spectral densities and raw time series for representative channels (Ch 0, Ch 4, Ch 7) are shown in Figure 19A for both conditions.

Geometric consequences. Figure 19A shows the lag embeddings recovered from individual channels under both conditions across five representative channels (Ch 1–5). Under the single source condition (left column), all channels recover clean toroidal or disk-like embeddings consistent with the predicted oscillatory manifold structure. Under volume conduction (right column), the embeddings collapse toward ellipsoidal or disk-like structures with degraded toroidal geometry, reflecting the loss of independent phase information across channels due to spatial mixing.

These results have two important implications. First, they confirm that clean toroidal geometry in single-channel recordings is a signature of genuine local oscillatory dynamics rather than a trivial consequence of spatial structure. Second, they show that volume conduction degrades but does not destroy geometric structure — the embeddings remain low-dimensional and bounded, consistent with the theoretical prediction that aperiodic and mixed components contribute thickness rather than new topological dimensions (Section 2).

K.2. Procrustes Disparity Diagnostic

A key practical challenge in applying our framework to real data is distinguishing genuine toroidal geometry from spurious torus fits that arise from noise or aperiodic structure. We address this using a Procrustes disparity diagnostic: a data-driven threshold calibrated on synthetic signals with known ground-truth geometry.

Diagnostic construction. We generate two classes of synthetic signals. *True torus* signals (class A) are drawn from AR processes with sustained oscillatory poles, producing genuine toroidal lag embeddings. *Spurious torus* signals (class B) are drawn from AR processes with only stable aperiodic poles, producing embeddings that may superficially resemble tori under

finite-sample noise but lack genuine circular topology. For each signal, we fit a torus to the lag embedding and compute the mean Procrustes disparity between the fitted torus and the point cloud — a measure of how well the torus model accounts for the observed geometry.

Threshold calibration. Figure 19B shows the resulting distributions of mean Procrustes disparity for true torus (dark red) and spurious torus (gray) signals. The two distributions are well-separated, with true torus signals concentrated near zero disparity and spurious torus signals exhibiting large, diffuse disparity values. A threshold of 0.0673 provides near-optimal separation between the two classes.

Classification performance. Figure 19C shows the confusion matrix for the Procrustes diagnostic classifier applied to held-out synthetic signals. The classifier achieves 93.3% accuracy on true torus signals and 100% accuracy on spurious torus signals, confirming that mean Procrustes disparity is a reliable and practically useful criterion for distinguishing genuine from spurious toroidal geometry.

K.3. Cross-Channel Validation in Monkey V1

Figure 19D shows lag embeddings recovered from four representative channels of the monkey V1 Utah array. Consistent with the single-source simulation results in Appendix K.1, all channels recover qualitatively similar toroidal geometry, confirming that the manifold structure reflects population-level oscillatory dynamics rather than channel-specific artifacts. The intrinsic dimension estimated from each channel is consistent with the two-torus prediction of Section 2, with optimal AR order concentrating near $p = 4$ across channels.

K.4. Manifold Visualization for Mice recordings

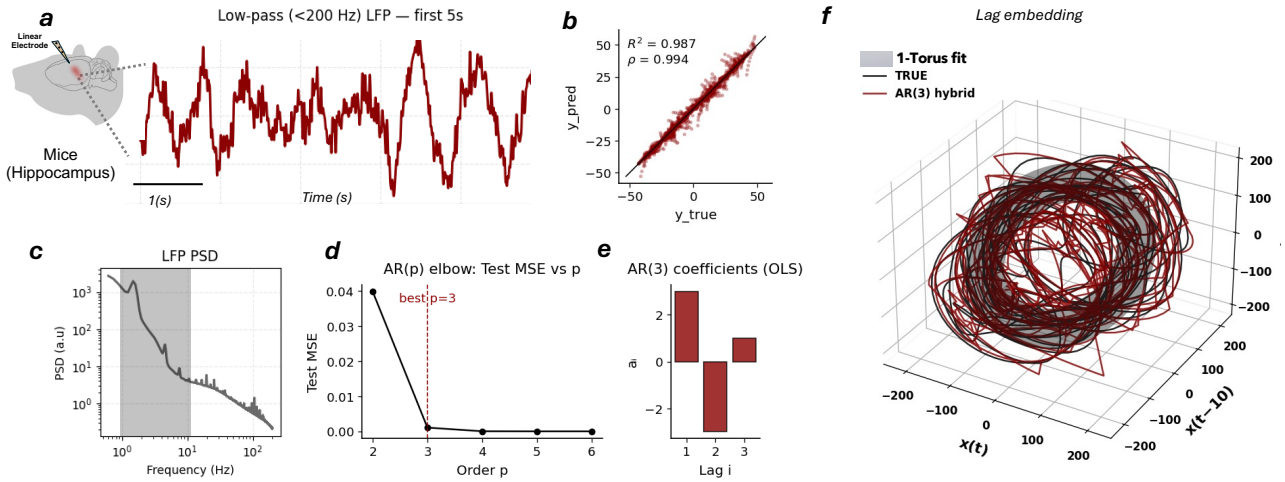


Figure 17. **Manifold reconstruction of mouse (hippocampus) LFP.** (a) Example segment of low-pass filtered (<200 Hz) local field potential recorded from mouse hippocampus. (b) Prediction accuracy of the fitted AR(p) model on held-out test data: scatter of predicted vs. true samples with R^2 and Pearson ρ confirming strong agreement. (c) Power spectral density (PSD) of the signal, exhibiting a dominant narrowband oscillatory component together with a broadband $1/f$-like background. (d) Model-order selection via the "elbow" in test MSE, yielding p=3 as the optimal order. (e) Estimated AR(3) coefficients obtained via ordinary least squares. (f) Delay-embedded trajectory of the LFP and the corresponding AR(p) free-run prediction, overlaid with the fitted 1-torus geometry. The embedding indicates that the hippocampal LFP is well described by a single circular factor, with broadband fluctuations primarily contributing thickness around the torus rather than altering its intrinsic topology.

K.5. Diagnostic Applied to Real Recordings

Figure 19E shows the Procrustes disparity diagnostic applied to real neural recordings, overlaid on the synthetic calibration distributions from Appendix K.2. Monkey V1 LFP recordings yield a mean Procrustes disparity of 0.1477 (blue vertical line), above the true torus threshold (0.0820, orange dashed line) but well below the bulk of the spurious torus distribution, indicating geometrically valid toroidal structure with moderate noise — consistent with the two dominant oscillatory modes and broadband background observed spectrally in Figure 4. Mouse cortex EEG recordings yield a mean Procrustes disparity

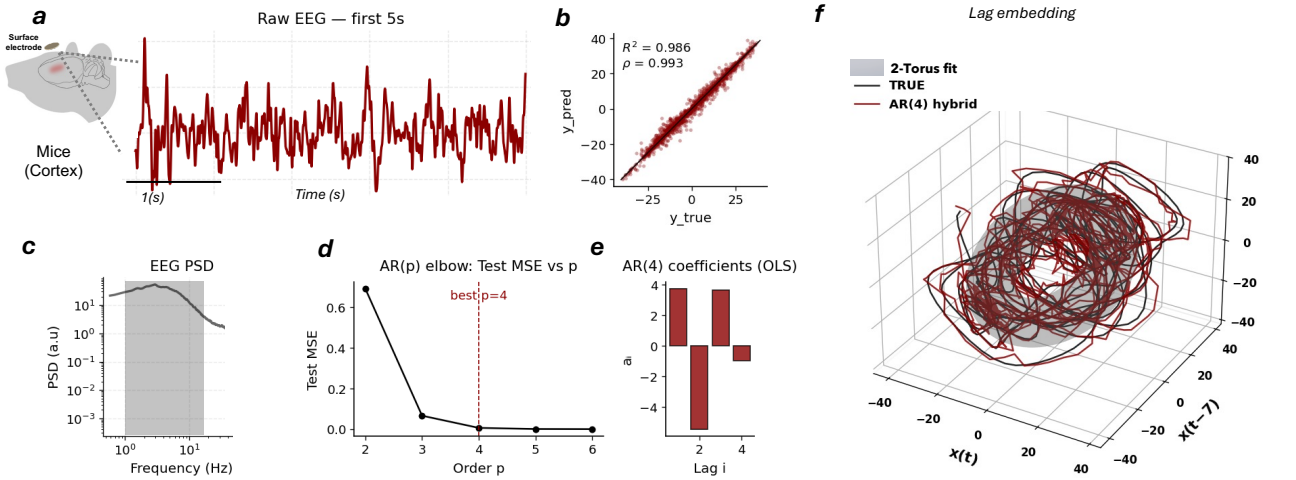


Figure 18. **Manifold reconstruction of mouse (cortex) EEG.** (a) Example segment of raw EEG signal recorded from mouse cortex via surface electrode, showing the first 5 s of broadband cortical activity. (b) Prediction accuracy of the fitted AR(p) model on held-out test data: scatter of predicted vs. true samples with $R^2 = 0.986$ and Pearson $\rho = 0.993$ confirming strong agreement. (c) Power spectral density (PSD) of the EEG signal, displaying a $1/f$-like spectral profile dominant in the low-frequency range with no prominent narrowband oscillatory peak. (d) Model-order selection via the "elbow" in test MSE, yielding p=4 as the optimal order. (e) Estimated AR(4) coefficients obtained via ordinary least squares, showing alternating sign structure characteristic of oscillatory dynamics. (f) Delay-embedded trajectory of the EEG and the corresponding AR(4) free-run prediction, overlaid with the fitted 2-torus geometry. Despite the absence of a dominant narrowband peak, the cortical EEG trajectory organizes onto a two-circle product manifold, suggesting that the interplay of multiple broadband oscillatory components is collectively captured by a toroidal delay embedding.

of 0.4092 (green vertical line), falling within the spurious torus distribution, reflecting the noisier and less geometrically clean torus structure observed qualitatively in Figure 18. Importantly, this does not indicate failure of the framework: rather, the diagnostic correctly grades geometric quality on a continuous scale, reflecting the degree to which the signal's oscillatory structure is sustained and spectrally distinct. The mouse EEG result is consistent with the absence of a dominant narrowband peak in its power spectrum (Figure 18c) and with the theoretical prediction that weaker or broader oscillatory modes yield thicker, less geometrically precise tori.

Summary. Together, these results establish three points. First, single-channel lag embeddings reflect genuine local oscillatory geometry rather than volume conduction artifacts. Second, the Procrustes disparity diagnostic reliably separates genuine from spurious toroidal geometry on synthetic data. Third, real LFP and EEG recordings fall at interpretable positions on the diagnostic scale, with monkey V1 LFP showing cleaner toroidal geometry than mouse cortex EEG, consistent with their respective spectral structures and the predictions of Section 2.

L. Windowed persistent homology of monkey V1 LFP

To assess the topological structure of macaque V1 LFP independently of the parametric torus fit, we apply a windowed Vietoris–Rips persistent homology pipeline to a delay-embedded single channel and report the empirical distribution of recovered Betti vectors across windows.

Let $x(t)$ denote the preprocessed LFP from a single channel sampled at $F_s = 500$ Hz over a 20 s recording. Preprocessing consists of 50 Hz notch filtering, 1–50 Hz bandpass filtering, and division by the Hilbert amplitude envelope computed in the 1–80 Hz band and low-pass smoothed at 3 Hz, followed by rescaling by the median envelope amplitude. We delay-embed $x(t)$ at lag τ into $\mathbb{R}^3$:

$$\Phi_\tau(t) = \big(x(t),\, x(t+\tau),\, x(t+2\tau)\big) \in \mathbb{R}^3, \tag{10}$$

with $\tau = 40$ samples (80 ms) (Takens, 2006; **?**). From the embedded trajectory we draw $N = 200$ random windows of length $L = 2000$ samples (4 s; seed $= 42$), sub-sample each window to $M = 300$ points, and project each point cloud onto the unit sphere by centering and dividing each point by its Euclidean norm. Let $\mathcal{Z}^{(n)} = \{z_i^{(n)}\}_{i=1}^{M} \subset S^2$ denote the n-th preprocessed cloud.

For each $\mathcal{Z}^{(n)}$ we compute the Vietoris–Rips persistence diagram up to dimension 2 with Ripser (Tralie et al., 2018), yielding a multiset of (birth, death) pairs in each dimension $k \in \{0, 1, 2\}$. We summarize the diagram by a Betti vector

$$\beta^{(n)} = \big(\beta_0^{(n)},\, \beta_1^{(n)},\, \beta_2^{(n)}\big), \tag{11}$$

where $\beta_0^{(n)} = 1$ by convention (counting the single infinite component of the filtration) and, for $k \in \{1, 2\}$,

$$\beta_k^{(n)} = \#\big\{(b, d) \in \mathrm{PD}_k\big(\mathcal{Z}^{(n)}\big) \,:\, d - b > \theta\big\}, \tag{12}$$

with persistence threshold $\theta = 0.30$. We report the empirical distribution

$$\hat{p}(\beta) = \frac{1}{N} \sum_{n=1}^{N} \mathbf{1}\big\{\beta^{(n)} = \beta\big\} \tag{13}$$

over the 200 windows.

Intuitively, β_1 counts the number of independent loops and β_2 the number of enclosed cavities in the embedded trajectory at the chosen scale, so the joint distribution $\hat{p}(\beta)$ summarizes the homotopy type recovered from V1 LFP across independently sampled windows. The reference signature for a 2-torus is $\beta = (1, 2, 1)$, and a concentration of mass near this value provides window-level, model-free evidence of toroidal structure that complements the parametric torus fit in Sec. **??**. Results depend on three discretization choices—the embedding lag τ, the joint pair (window length, sub-sample size), and the persistence threshold θ—over which we report sensitivity sweeps in Sec. E.

M. Torus score and torus error

To quantify the quality of the recovered toroidal geometry, we report two measures: *torus score* and *torus error*. For each recording window, let $\mathcal{Z} = \{z_i\}_{i=1}^{N}$ denote the lag-embedded trajectory, with $z_i \in \mathbb{R}^q$. We fit the toroidal geometry predicted by the recovered dynamics and denote its fitted geometric region by $\widehat{\mathcal{T}}$. This region is allowed to have finite thickness, since empirical trajectories are not expected to lie exactly on an ideal torus in the presence of noise, broadband fluctuations, and finite-sample variability.

For each embedded point, we define a nonnegative residual

$$e_i = \begin{cases} 0, & z_i \in \widehat{\mathcal{T}}, \\ \min\limits_{u \in \partial\widehat{\mathcal{T}}} \|z_i - u\|_2, & z_i \notin \widehat{\mathcal{T}}, \end{cases} \tag{14}$$

where $\partial\widehat{\mathcal{T}}$ denotes the boundary of the fitted torus region. Thus, points already contained by the recovered toroidal geometry contribute zero residual, while points outside the fitted region contribute their shortest residual distance back to the torus boundary. The *torus score* is the fraction of embedded points contained by the fitted torus,

$$\text{Torus Score} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{e_i = 0\}, \tag{15}$$

and the *torus error* is the mean residual distance,

$$\text{Torus Error} = \frac{1}{N}\sum_{i=1}^{N} e_i. \tag{16}$$

Intuitively, torus score measures how much of the lag-embedded trajectory is captured by the recovered toroidal geometry, whereas torus error measures the average distance by which the remaining points fall outside that geometry. A high torus score together with a low torus error indicates that the trajectory is well explained by the fitted torus. Torus error is reported in the units of the embedding coordinates and is interpreted under the same preprocessing and normalization used for the corresponding embedding. In practice, the torus fit is initialized from the principal axes of the embedded point cloud and optimized with mild geometric regularization to avoid degenerate solutions, such as collapsed holes or unrealistically large radii.

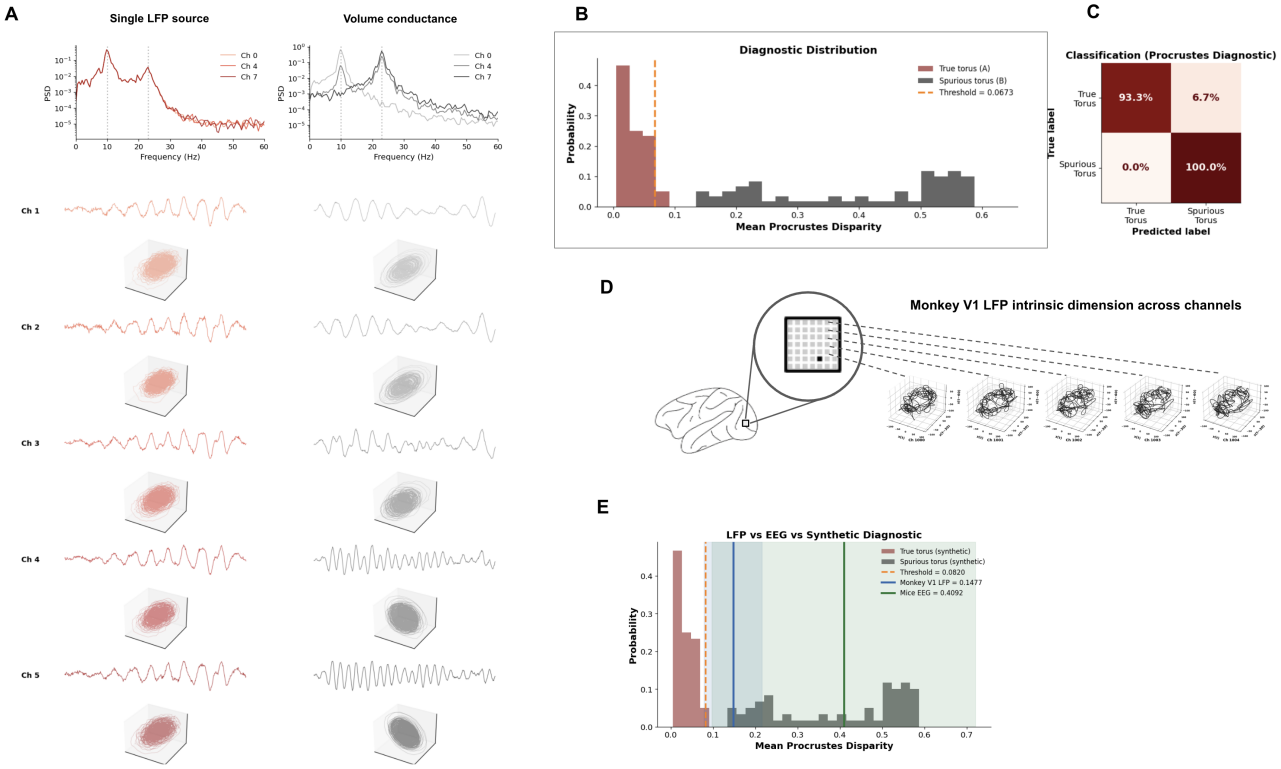


Figure 19. **Multichannel analysis and Procrustes disparity diagnostic. (A)** Comparison of single LFP source (left) versus volume conduction (right) conditions. Top: power spectral densities for representative channels (Ch 0, Ch 4, Ch 7) showing similar spectral profiles across conditions. Bottom: lag embeddings recovered from five individual channels (Ch 1–5) under each condition. Single source recordings yield clean toroidal or disk-like embeddings (red), while volume-conducted signals produce degraded ellipsoidal geometry (gray), confirming that toroidal structure reflects genuine local oscillatory dynamics. **(B)** Distributions of mean Procrustes disparity for synthetic true torus signals (dark red) and spurious torus signals (gray). The two distributions are well-separated, with a diagnostic threshold of 0.0673 (orange dashed line) providing near-optimal classification. **(C)** Confusion matrix for the Procrustes disparity classifier on held-out synthetic signals. The classifier achieves 93.3% accuracy on true torus signals and 100% accuracy on spurious torus signals. **(D)** Cross-channel validation in monkey V1. Lag embeddings recovered from four representative channels of the Utah array (Ch 1001–1004) show consistent two-torus geometry across channels, confirming that the recovered manifold structure reflects population-level dynamics rather than channel-specific artifacts. **(E)** Procrustes disparity diagnostic applied to real recordings, overlaid on synthetic calibration distributions. Monkey V1 LFP (blue line; disparity = 0.1477) falls above the true torus threshold (0.0820, orange dashed) but below the spurious torus bulk, indicating valid toroidal geometry with moderate noise. Mouse cortex EEG (green line; disparity = 0.4092) falls within the spurious torus distribution, consistent with its noisier spectral structure and the absence of a dominant narrowband oscillatory peak. The diagnostic correctly grades geometric quality on a continuous scale rather than producing binary classifications.